



ISE 726: Systems Modeling

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(ISEEM)

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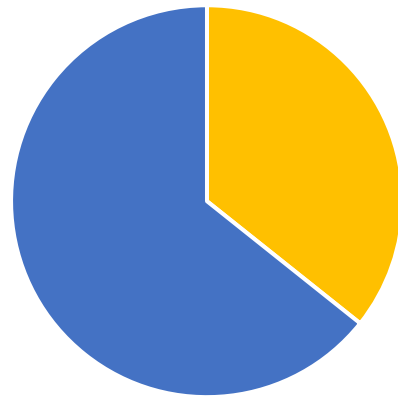
- To comply with federal Title IV financial aid requirements, UAH must verify that students have begun attending or academically participating in each course for which they are registered.
- For this reason, all students must demonstrate participation during the first **10 days of class**. Participation may include attending class, submitting an assignment, completing a quiz or exam, or participating in another academically related activity.
- Please note that simply being registered for the course does not count as attendance. If a student does not begin attending or participating, this may affect financial aid eligibility.

Learning Objectives

- **Introductions**
 - Personal Introductions
 - Course Introduction, incl. Learning Objectives
- Systems Engineering (SE) Overview
 - A bit of history
 - The “V”-Model
 - SE Standards and Handbooks
 - Challenges of current practice
- Stakeholder Analysis
 - Identifying Stakeholders
 - CONOPS
 - Stakeholder Value Network (SVN) Analysis

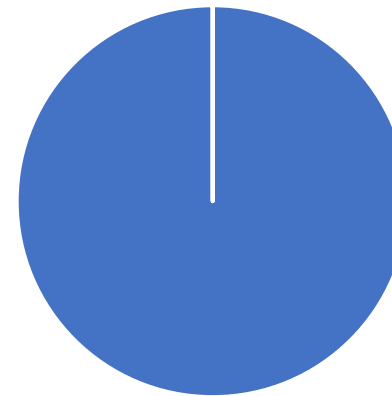
Grades Distributions:

9 – As and 5 - Bs



■ F ■ D ■ C ■ B ■ A

25 A(s)



■ F ■ D ■ C ■ B ■ A

Reciprocal Interview:

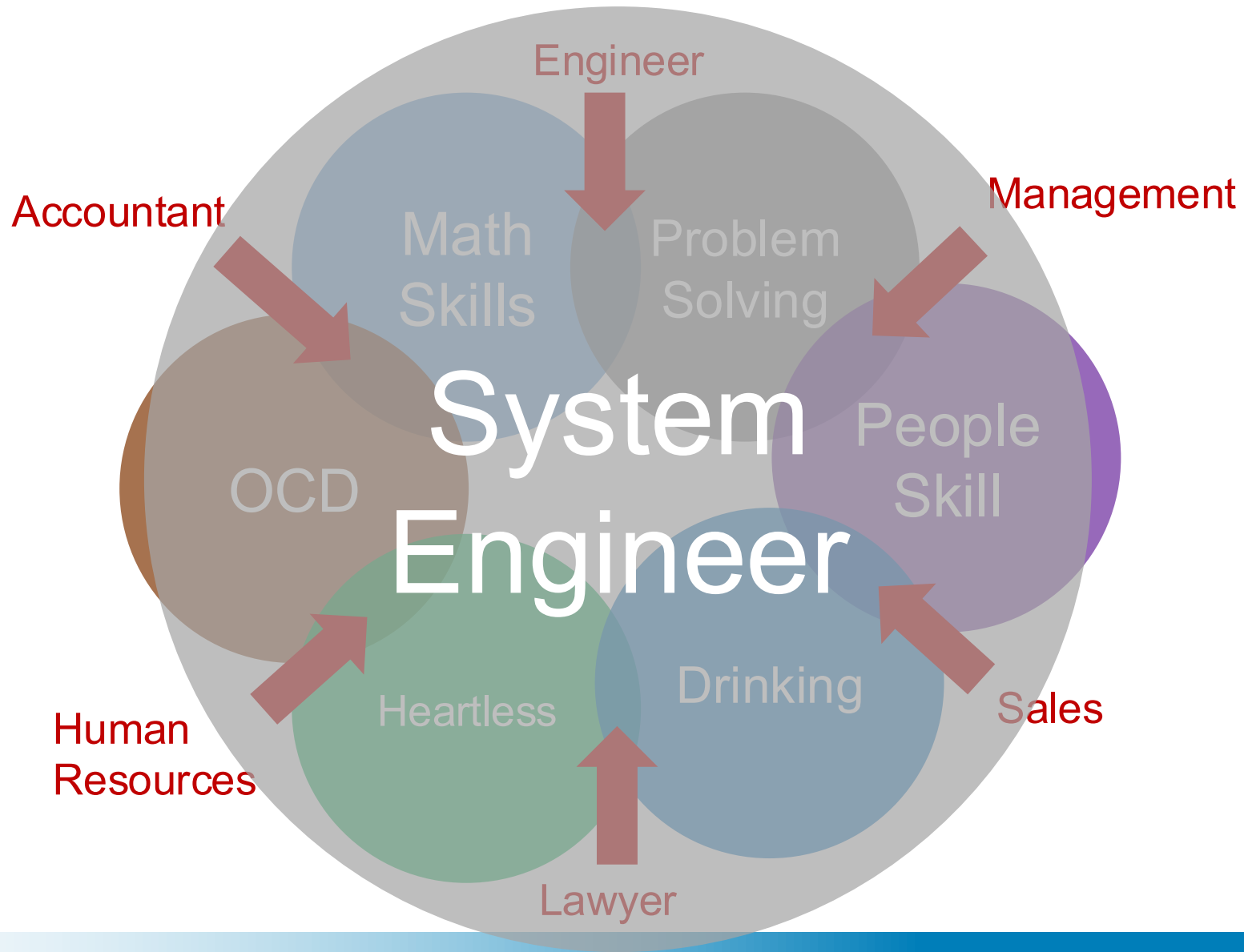
What excited you about this class?

What are your goals for this course?

What can I do to help you throughout this semester?

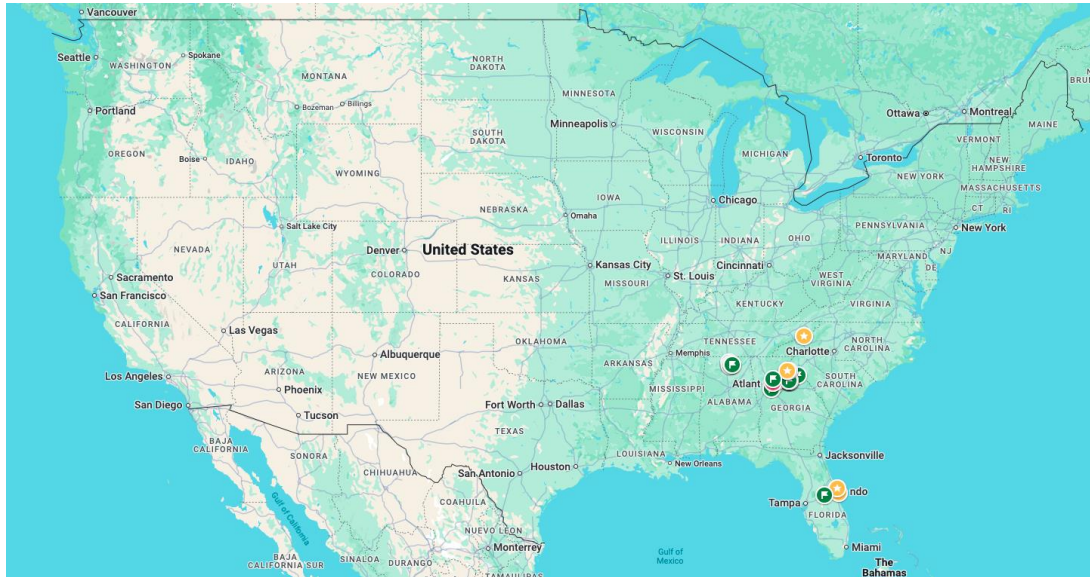
What frightens you about this class?

Why become an engineer? – Venn diagram



About the instructor:

Where I've lived:



Hobbies:



Education:



Ph.D. in Engineering



M.S., in Aerospace Engineering



B.S., in Mechanical Engineering

Self-Introduction

- Your name, department/lab affiliation, and research interests
- Any prior experience with system engineering? Any programming experience?
- Name one thing you want to learn in this class

Let's review the syllabus together.

Email Etiquette:

- When email a teacher, ALWAYS include your **full name and course**
- Include your class and what the email is specifically regarding in the **subject**
- Example:

Subject: ISE 430: Question about Homework 3

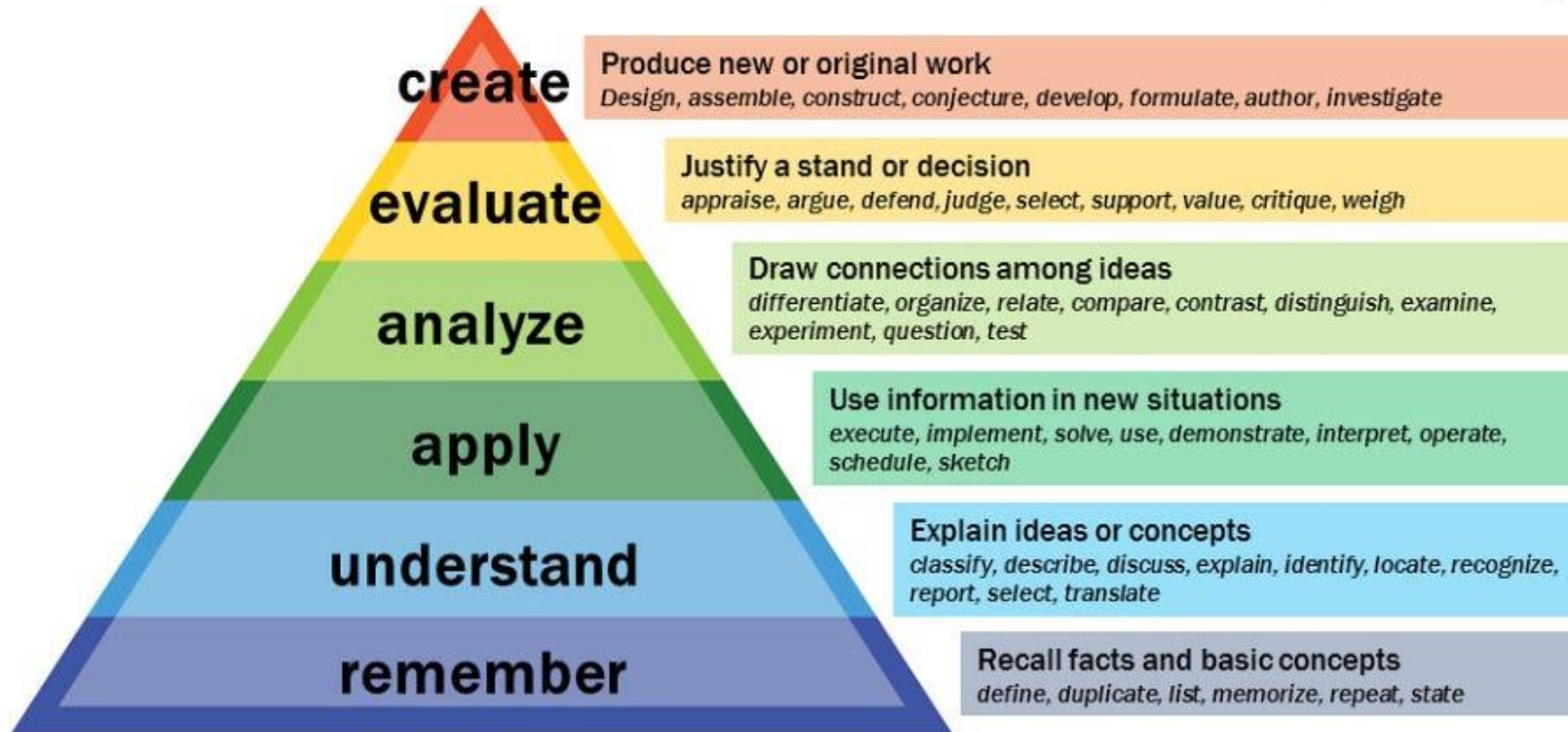
Additional Notes:

- Unless otherwise instructed, each student can collaborate on assignments **together**. However, discussion (not copying) of assignments with classmates is encouraged.
- Students are expected to keep the neatness and professionalism on all items submitted in the course. **Marks will be deducted** if professionalism is lacking, e.g., homework is not identified, questions are answered out-of-order, papers are stapled out-of-order, units are not specified, etc.

Syllabus!

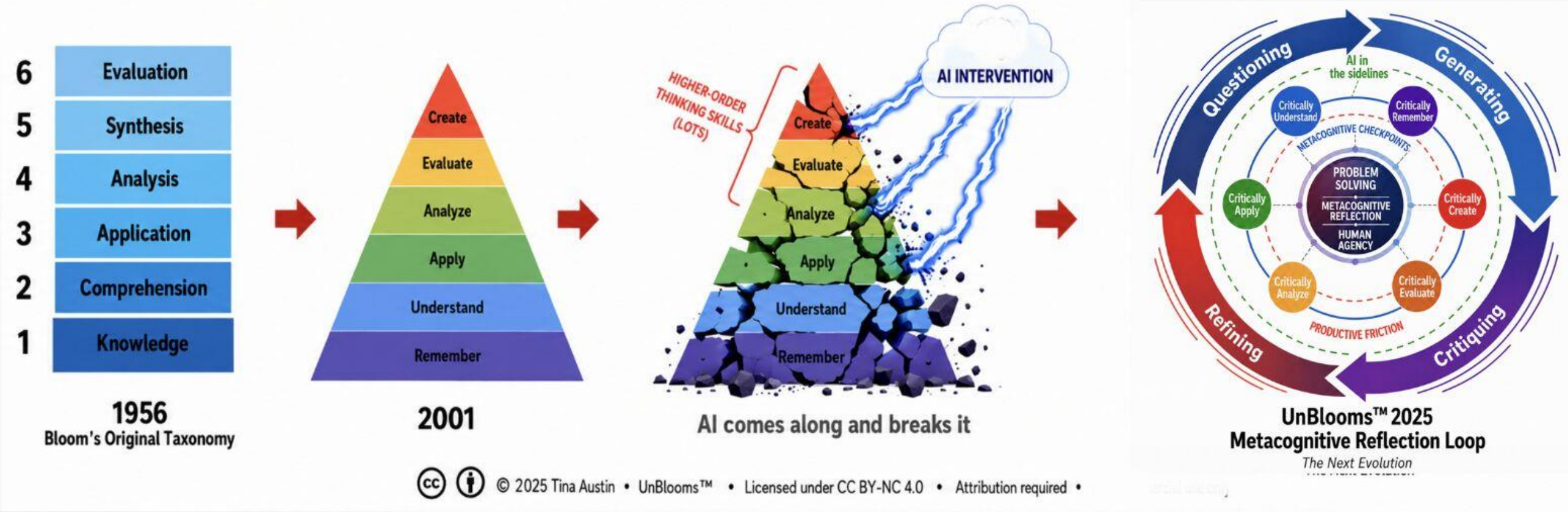
How to learn anything? (1/3)

Bloom's Taxonomy



How to learn anything? (2/3)

The Evolution of Bloom's Taxonomy to UnBlooms™



How to learn anything? (3/3)

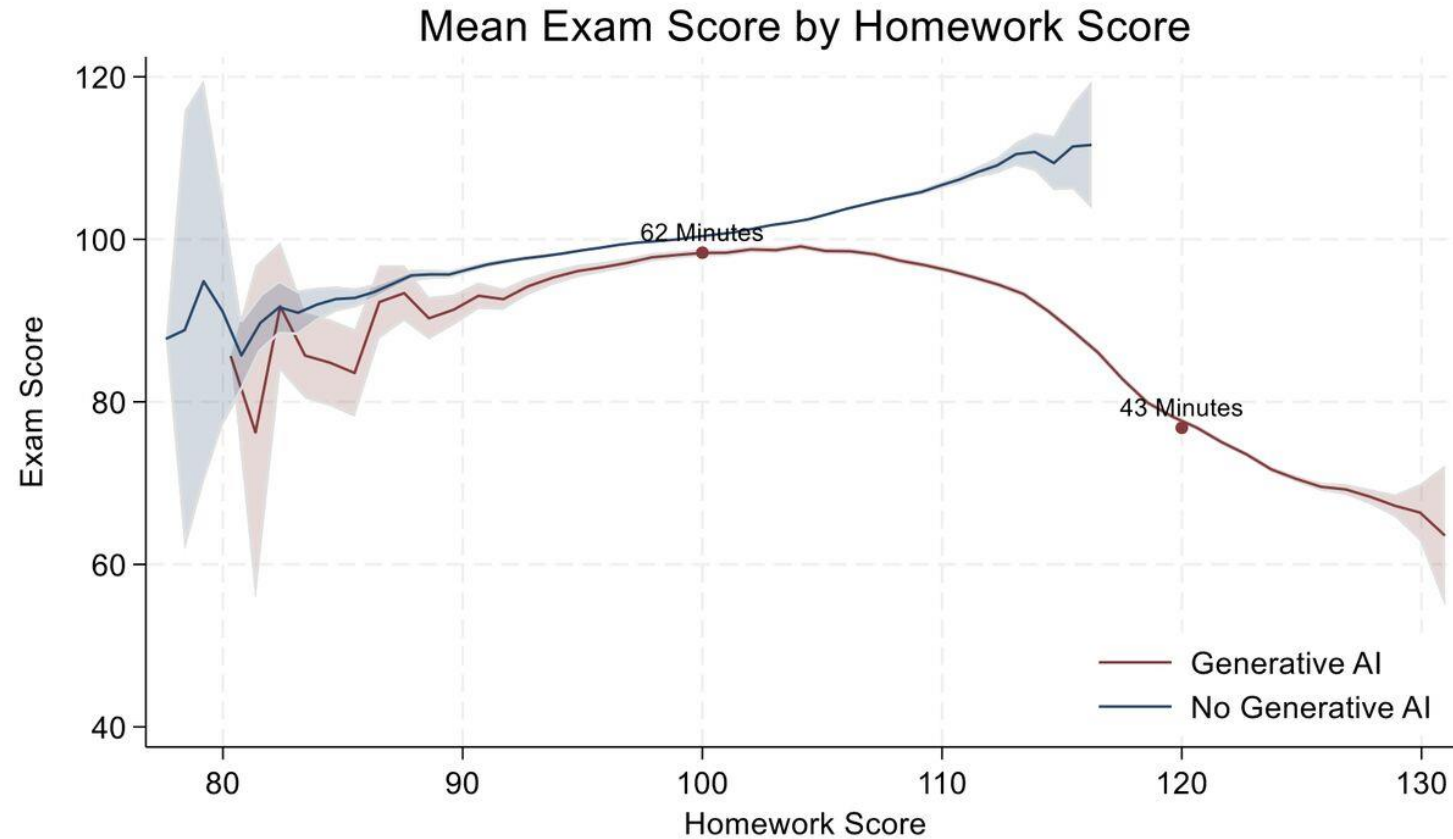
How does UnBlooms™ make a difference in learning?

Aspect	Traditional Bloom's	UnBlooms™
Starting point	Teacher delivers content → students absorb	Students generate initial understanding → test it against AI
AI's role	Threat to academic integrity OR ignored entirely	Partner for thinking <i>with</i> and <i>against</i>
Learning focus	Mastering enzyme facts and functions	Understanding <i>how we know</i> what we know about enzymes
Student thinking	Linear: memorize → apply → create	Recursive: predict → compare → revise → interrogate
Evidence of learning	Correct final answer/report	Visible reasoning process and metacognitive awareness
What's valued	Getting it right	Understanding why you were wrong and what that reveals
Failure is...	Something to hide	Data for learning

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What can you tell?

Figure A11. Homework and Exam Scores

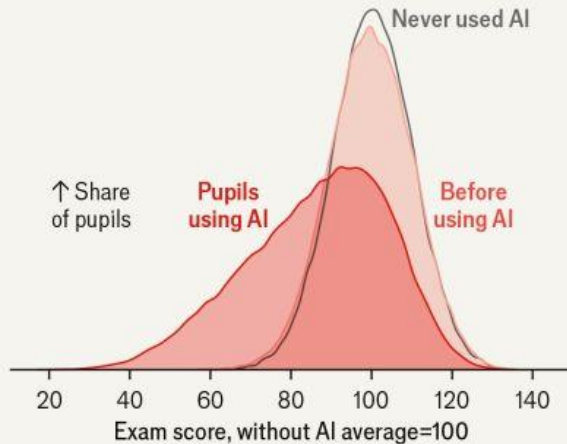


Note: The figure plots the smoothed mean and its 95% confidence intervals, by homework score, separately for students who have adopted generative AI and those who have not. The data also shows the average homework completion time for AI students with homework scores of 70 and 90. The data is at the student-month level and the sample summary statistics are shown in Panel A of Table 1.

What can you tell?

...but come exam time the technology proves less helpful

China*, Jan 2023-Jun 2025

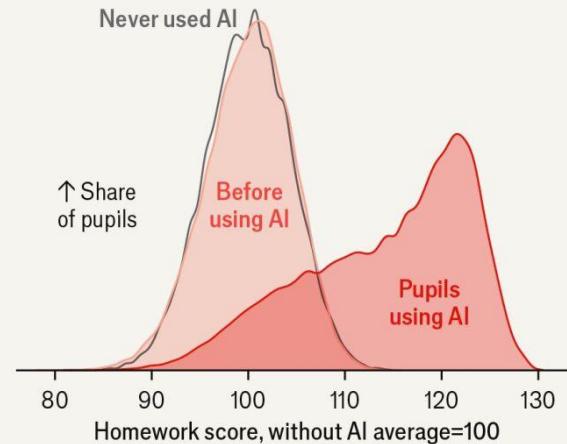


*26,811 pupils aged 12-18
Source: "The generative AI learning penalty",
by D. Strömberg et al., preprint June 2026

E

Pupils using AI have higher average homework scores...

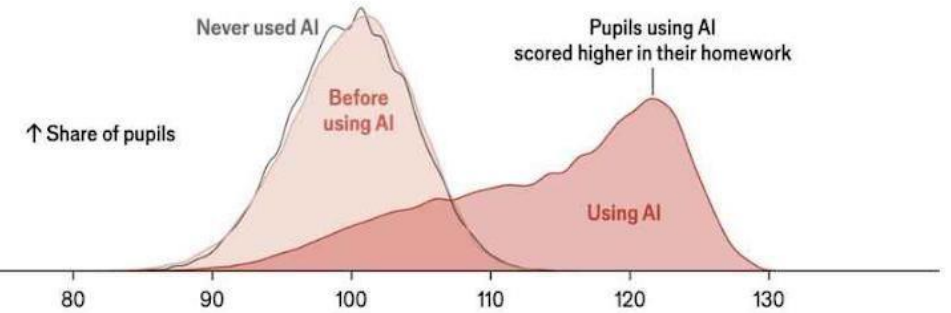
China*, Jan 2023-Jun 2025



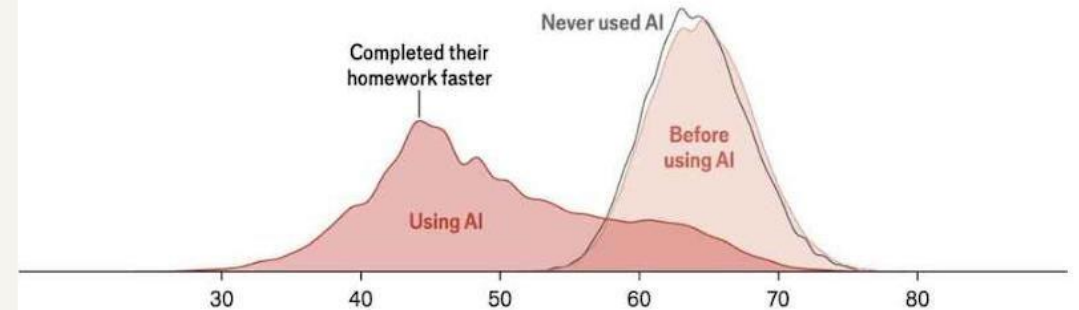
*26,811 pupils aged 12-18
Source: "The generative AI learning penalty",
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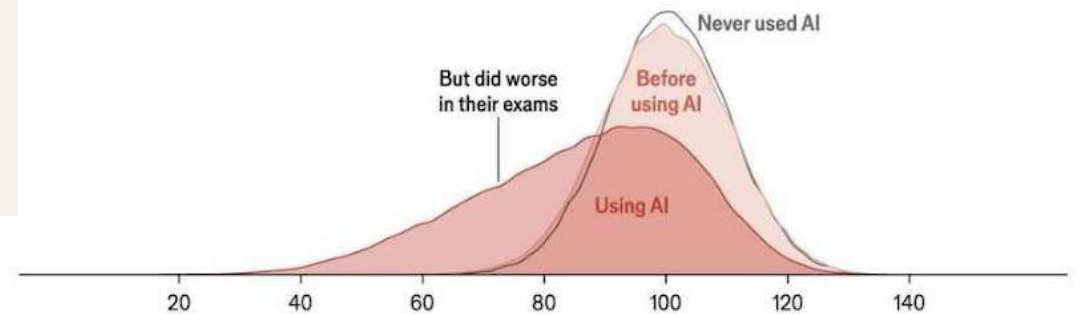
Homework score, average without AI=100



Homework completion time, minutes



Exam score, average without AI=100



Source: "The generative AI learning penalty: evidence from
Chinese secondary education", by D. Strömberg et al., preprint June 2026

*26,811 pupils aged 12-18

Expectations

Lectures

- 13 lectures total (2h each)
- Full-time and part-time students are expected to meet the same learning objectives
- Active engagement in class is expected (brownie cookie points for participation)

Assignments

- 4 assignments total (~2 weeks each)
- You are expected to practice coding on your own
- Strong self-learning ability is required (e.g., learning new tools or coding independently)
- I cannot help you debug your code

Readings

- Pre-readings based on sections of the NASA SE Handbook
- Post-readings: 1-2 journal or conference papers per week
- Expect to read ~30–40 pages per week
- Excellent learners take notes and self-organize knowledge across readings and lectures

Final Project

- Project proposal (Max 2 pages)
- 10-minute presentation
- Written final report

- System modeling is rapidly evolving
- Shifting from document-centric methods to Model-Based Systems Engineering (MBSE) for complex systems
- Imminent changes will place much greater emphasis on AI- and data-driven approaches, particularly for ill-defined systems and cyber-physical systems.

Machine Learning Systems Engineer
CentML • United States • via Indeed

Machine Learning Systems Engineer (1 Year Fixed Term)

Systems Research Engineer, Machine Learning Systems
Together AI • San Francisco, CA • via ZipRecruiter
\$ 160K-230K a year • Full-time • Health insurance

Artificial Intelligence Systems Engineer (DOD & Service Center)
The MITRE Corporation • Quantico, VA • via MITRE Jobs - MITRE Corporation
29 days ago • Full-time

Job highlights
Identified by Google from the original job post

Qualifications

- Strong background in machine learning, efficient inference for large language models
- Knowledge of ML/AI applications and as large language models and diffusion models and how they are used
- Knowledge of system performance and system architecture

2 more items(s)

Benefits

- We offer competitive compensation, equity, and benefits, as well as flexibility in terms of work
- The US base salary range for this full equity + benefits
- Our salary ranges are determined by experience and location

Responsibilities

- As a Systems Research Engineer you will play a crucial role in researching and developing new machine learning systems at Together AI
- Working closely with the modeling, AI design large-scale distributed training inference engine that serves a diverse set of customers
- Your research skills will be vital in staying at the forefront of innovation in machine learning systems

Job highlights
Identified by Google from the original job post

Qualifications

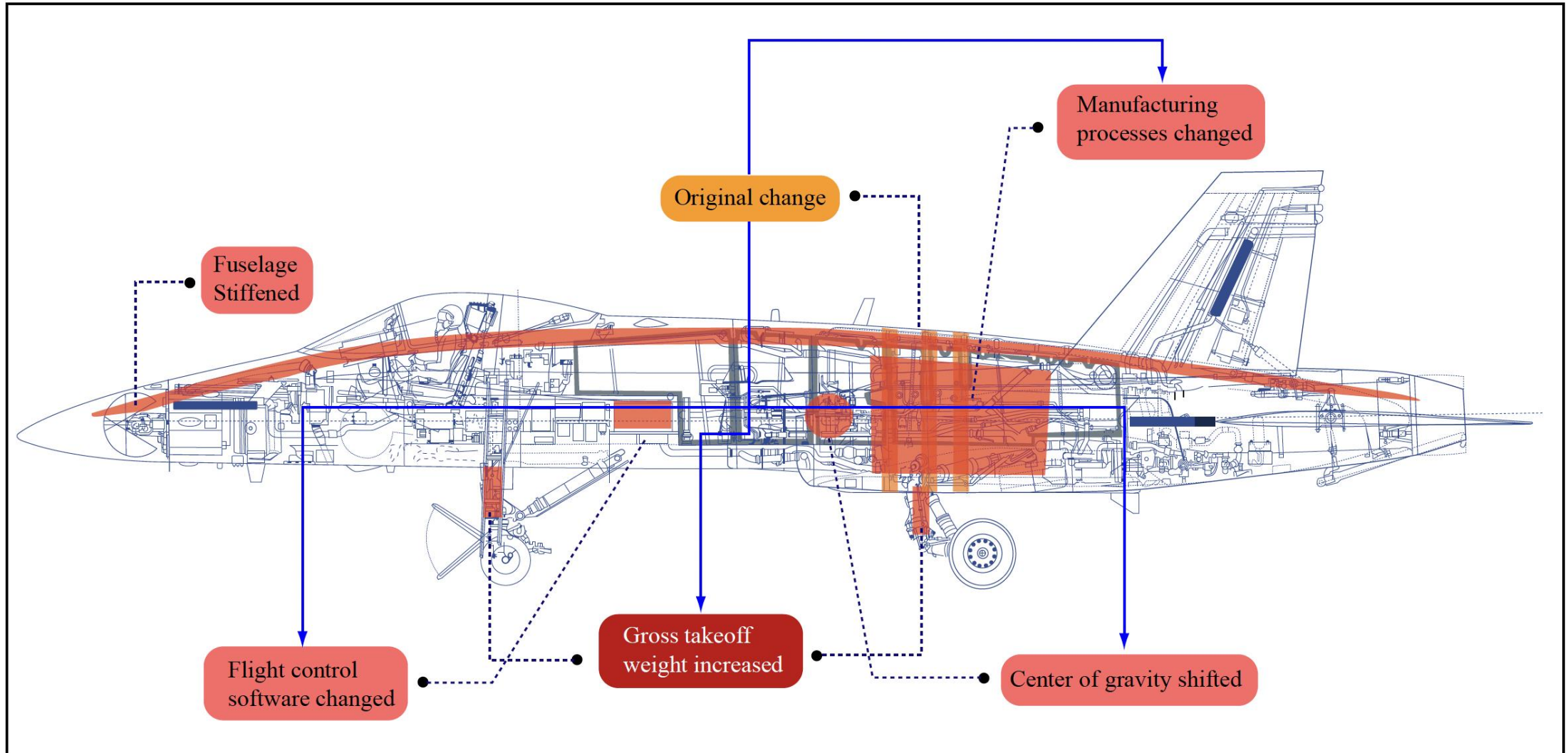
- The ideal candidate will have deep operational and domain knowledge of the Marine Corps meshed with an exquisite knowledge of current and future naval domain operational visions, doctrines, and tactical employments in the presence of US peer competitors
- The ideal candidate will be a champion for cultural and organizational transformation in data analytics, machine learning technologies, and the influence of artificial intelligence capabilities on society, industry, and defense
- Typically, interdisciplinary formal education background with at least a Bachelor of Science in a computer science field of study and 5 years of related experience; or 3 years and a Master's degree;

5 more items(s)

Benefits

- Salary compensation range and midpoint: \$142,000 - \$177,500 - \$213,000 Annual

F/A-18 Complex System Change



F/A-18 Complex System Change

- High-performance aircraft are very complex internally ... propulsion, avionics, structures ...
- Changing requirements can have ripple effects because everything is tightly coupled
 - It is difficult to predict the totality of system interactions ahead of time
- The “whole” system is much more than the air vehicle: logistics, training, incl. simulators etc..
- People matter a lot: contracts, culture, incentives

How would you define Systems Engineering?

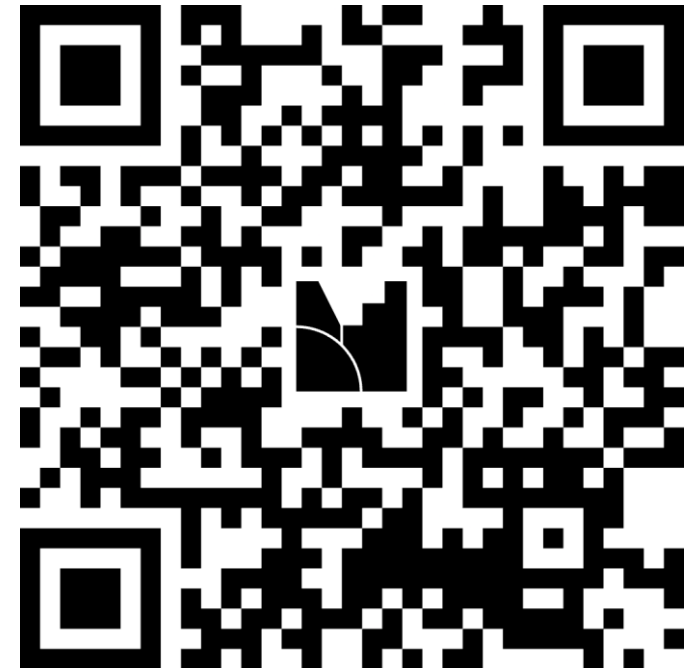
What is system modeling?

- Go to www.menti.com
- Enter code 7234 7177



How will system modeling shape future engineering practice?

- Go to www.menti.com
- Enter code 6980 9926



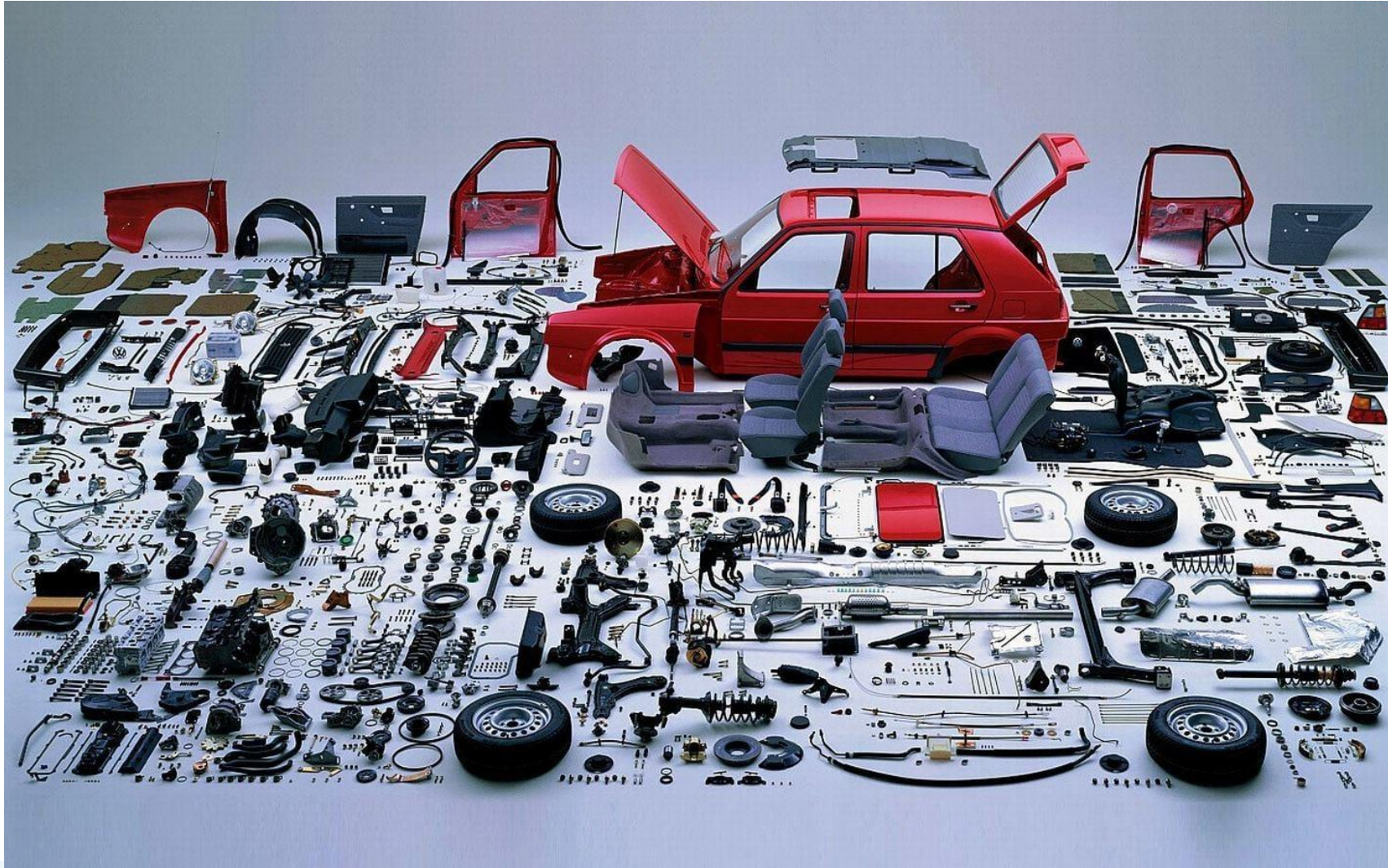
How would you define Systems Engineering?

- **Turn to your neighbor** and discuss for about **5 minutes**:
 - ☐ What is your definition of Systems Engineering?
 - ☐ Can you agree amongst yourselves?
 - ☐ What are the key elements of a definition?
- We will sample after about 5 minutes !

Motivation for this class

- Aerospace Systems deliver important functions to society ... air transportation, defense, sensing, exploration ...
- Complex “machines” with thousands of unique parts and potentially millions of interactions
 - Many aerospace systems require 6+ levels of decomposition to arrive at indivisible parts that cannot be taken “a-part”
- Humans play an important role as designers, operators, beneficiaries, maintainers
- Best Practices have emerged since the 1960’s and are continuously evolving ... documented in standards/handbooks
- Limitations of “traditional” SE
 - **System safety** ... recent SpaceX Falcon 9 launch failure
 - **Typical program cost and schedule overruns** ... Boeing Dreamliner 787 delays ...
- Systems Engineering is also penetrating in other industries
 - Automobiles, Software, Medical Devices

Why do we need system decomposition?



[Image Source:](#)

System Complexity

Assume 7-tree [Miller 1956]
<http://www.musanim.com/miller1956/>

- How many levels in drawing tree?

$$\#levels = \frac{\log(\#parts)}{\log(7)}$$

		#parts	#levels
•	Screwdriver (B&D)	3	1
•	Roller Blades (Bauer)	30	2
•	Inkjet Printer (HP)	300	3
•	Copy Machine (Xerox)	2,000	4
•	Automobile (GM)	10,000	5
•	Airliner (Boeing)	100,000	6

Simple



complex

Source: Ulrich, K.T., Eppinger S.D. , Product Design and Development
Second Edition, McGraw Hill, 2nd edition, 2000, Exhibit 1-3

Concept Question:

- How many levels of decomposition (depth of drawing tree) do we need to describe the car shown in the previous picture?
- 1
- 2
- 3
- 4
- 5
- 6
- >6
- This question does not make sense to me

Learning Objectives

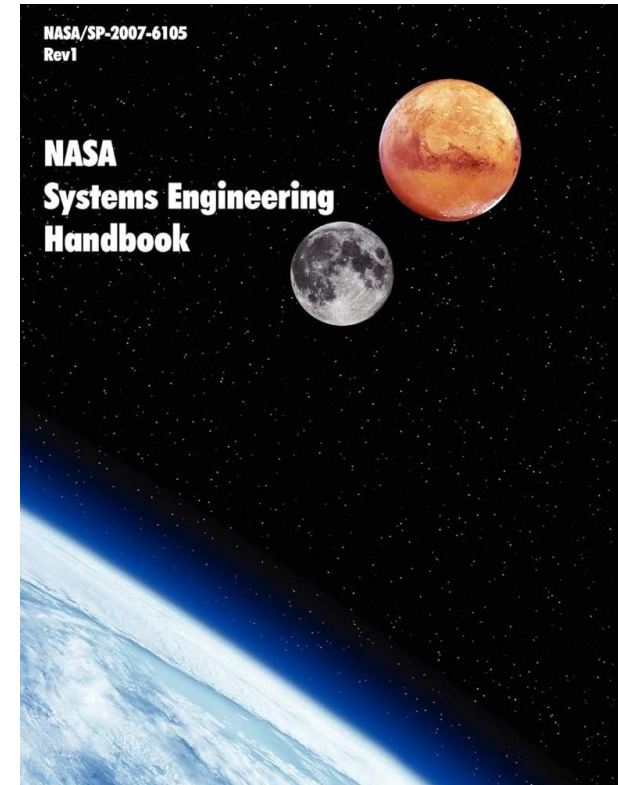
Participants in this class will be able to ...

- SE1: Describe the most important **Systems Engineering standards and best practices** as well as newly emerging approaches [1]
- SE2: Structure the **key steps in the systems engineering process** starting with stakeholder analysis and ending with transitioning systems to operations
- SE3: Analyze the **important role of humans** as beneficiaries, designers, operators and maintainers of aerospace and other systems
- SE4: Characterize the **limitations of the way that current systems engineering is** practiced in terms of dealing with complexity, lifecycle uncertainty and other factors
- SE5: **Apply some of the fundamental methods and tools** of systems engineering to a ‘simple’ cyber-electro-mechanical system as a stepping stone to more complex and real-world projects

[1] Our main “textbook” for the class will be the NASA Systems Engineering Handbook, NASA/TP-2007-6105, Rev 1.

NASA/SP-2007- 6105

- “Bible” for Systems Engineering at NASA
- Makes The Bridge From “Typical” Guidance Back To NASA Systems Engineering Process (NPR 7123.1)
 - Guidance From Practitioners
 - Written by practitioners for practitioners
 - “How” Vs “What”
- Updates The Guidance from SP-6105 (basic)
 - Updates The Practice/Methodology from 1995
- Provides Top-level Guidance for Systems Engineering Best Practices; It Is Not Intended In Any Way To Be A Directive
- Adds Additional Special Topics
 - Tools
 - NEPA
 - Human Factors



Learning Objectives

- Introductions
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- **Systems Engineering (SE) Overview**
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A bit of SE History

- Systems Engineering has been informally practiced since antiquity
 - Great Wall of China, Egyptian Pyramids, Roman Aqueducts
 - Mainly a “workforce” problem to build large infrastructures
- The term “**Systems Engineering**” can be traced back to Bell Labs (1940s)
 - https://en.wikipedia.org/wiki/Bell_Labs
 - Beginning of new methods to better handle complexity
- Formal Systems Engineering really started after WWII
 - 1950’s and 1960s: Cold War, Apollo Lunar Program, ICBMs etc...
 - Complex Engineering Systems: Air Traffic Control, High Speed Rail, Nuclear ..
 - Mainly (paper) document-based: requirements, specifications, test plans etc...
- Early Pioneers
 - Arthur D. Hall, Kelly Johnson, Simon Ramo, Eberhard Rechtin, Andrew Sage, Margaret Hamilton, and others
- 1995 Founding of International Council for Systems Engineering (INCOSE)
- Since ~2000: Development of new Model-Based-Systems-Engineering (MBSE). Need to accelerate SE and better handle complexity



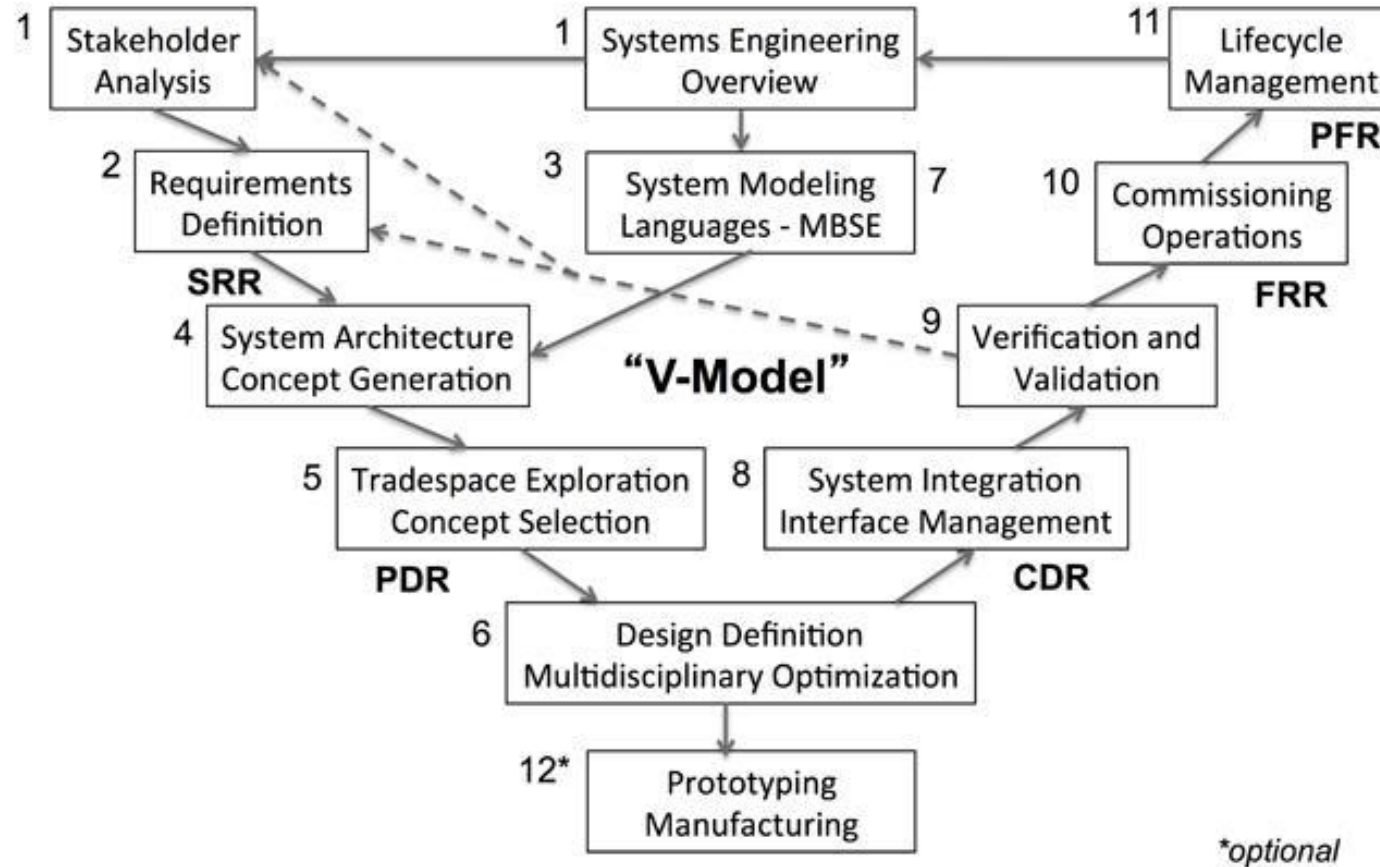
Simon Ramo

Some Definitions of SE

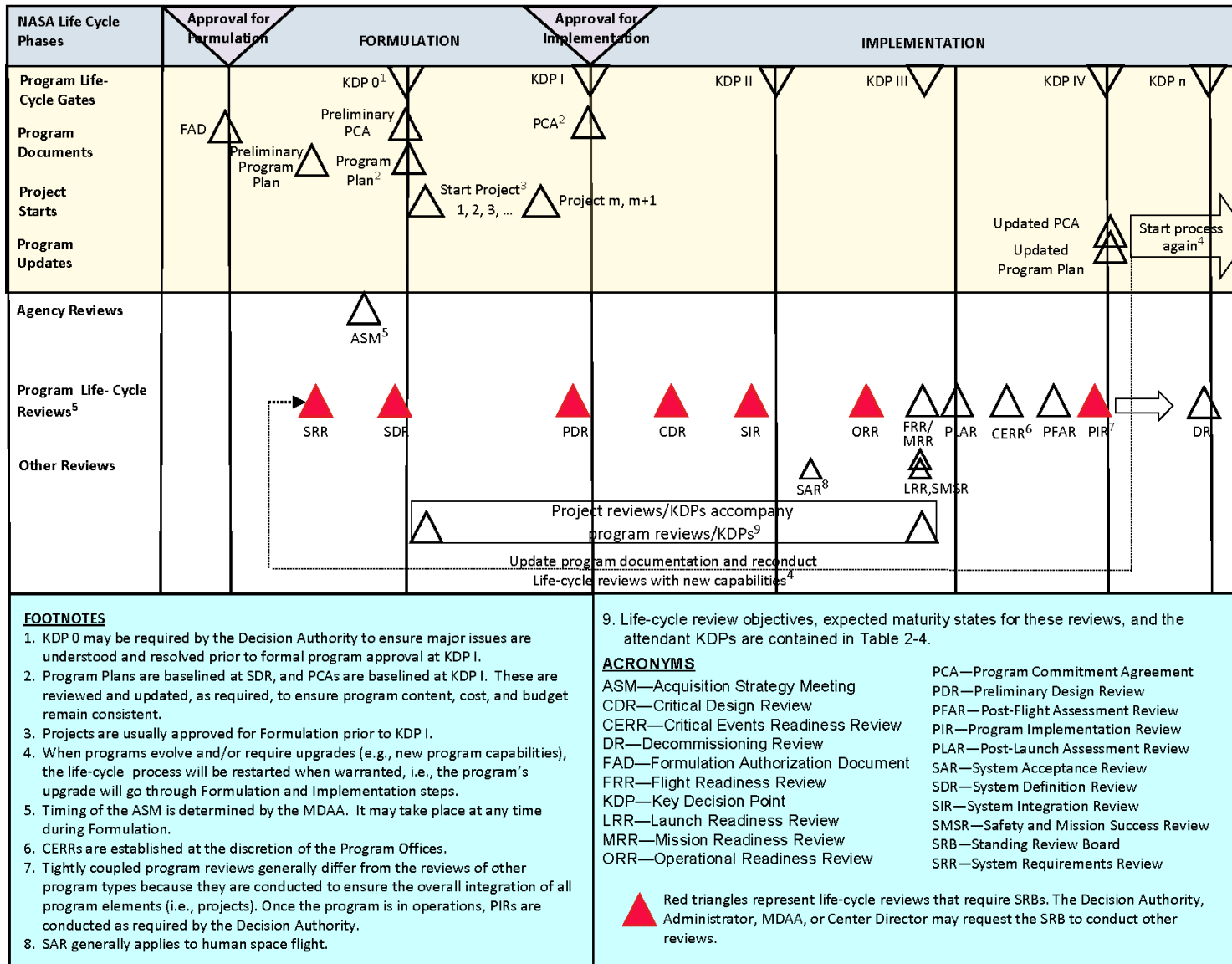
- "System engineering is a robust approach to the design, creation, and operation of systems. In simple terms, the approach consists of identification and quantification of system goals, creation of alternative system design concepts, performance of design trades, selection and implementation of the best design, verification that the design is properly built and integrated, and post-implementation assessment of how well the system meets (or met) the goals."- ***NASA Systems Engineering Handbook, 1995.***
- "An interdisciplinary approach and means to enable the realization of successful systems"- ***INCOSE handbook, 2004***
- More recently the scope of SE has broadened:
 - Design of Enterprises, Infrastructure Networks etc...
- ***AI4SE or SE4AI:***
 - How do we leverage AI-enabled capabilities to improve SE execution? (*AI4SE*)
 - How do we execute SE activities on AI-enabled systems? (*SE4AI*)



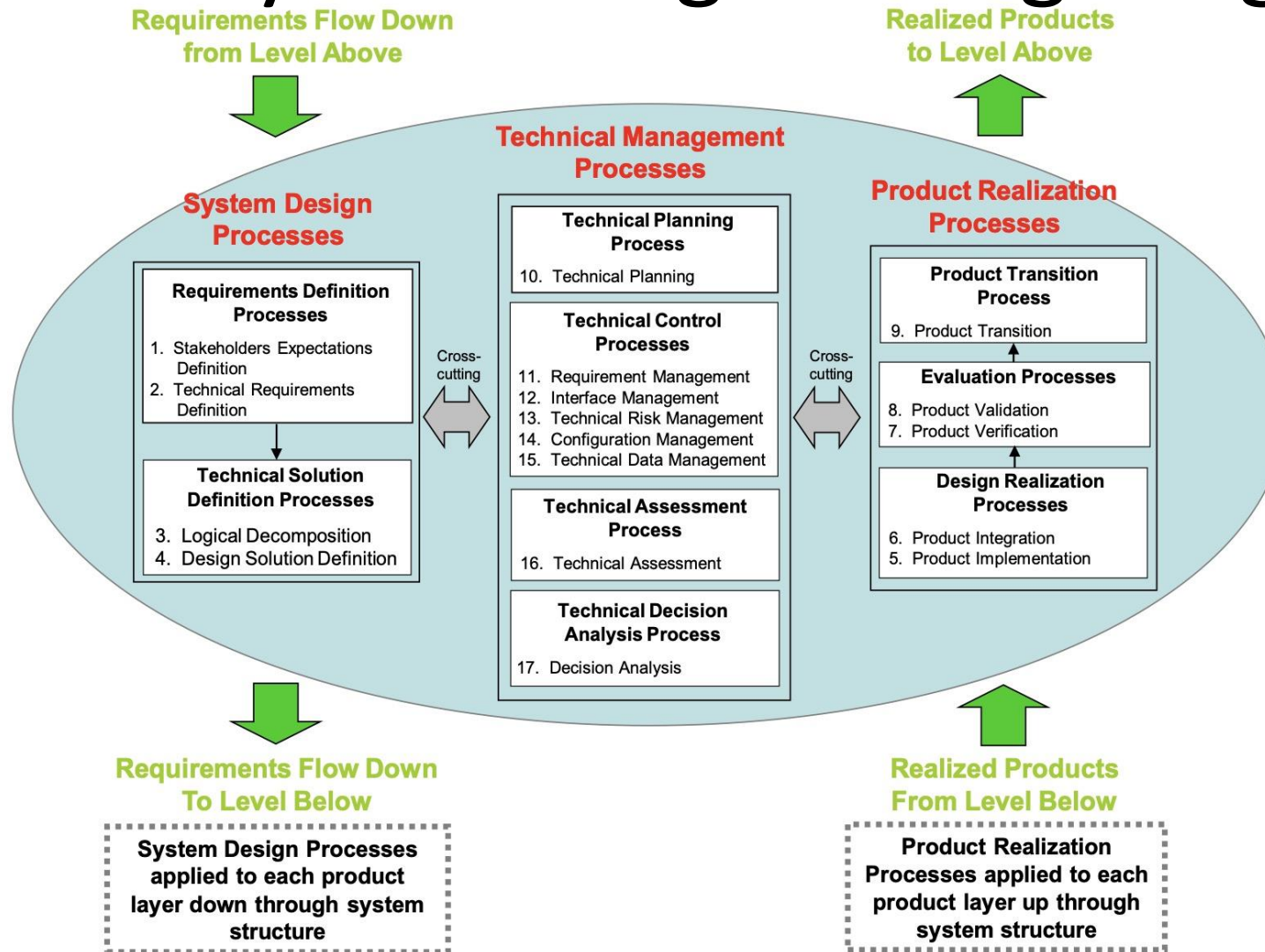
The famous “V-Model” of Systems Engineering



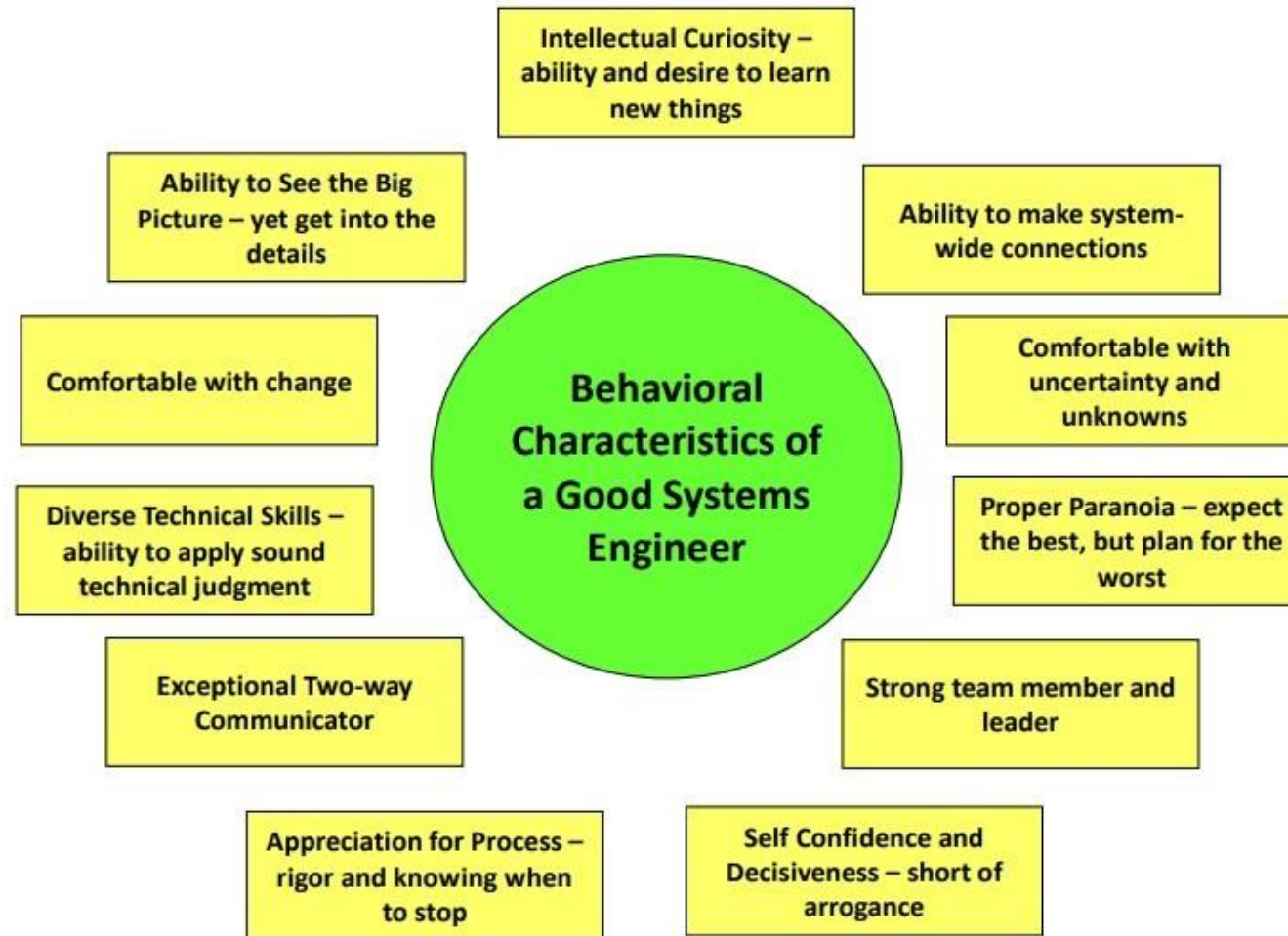
NASA Program & Project Lifecycle



The NASA Systems Engineering “Engine”



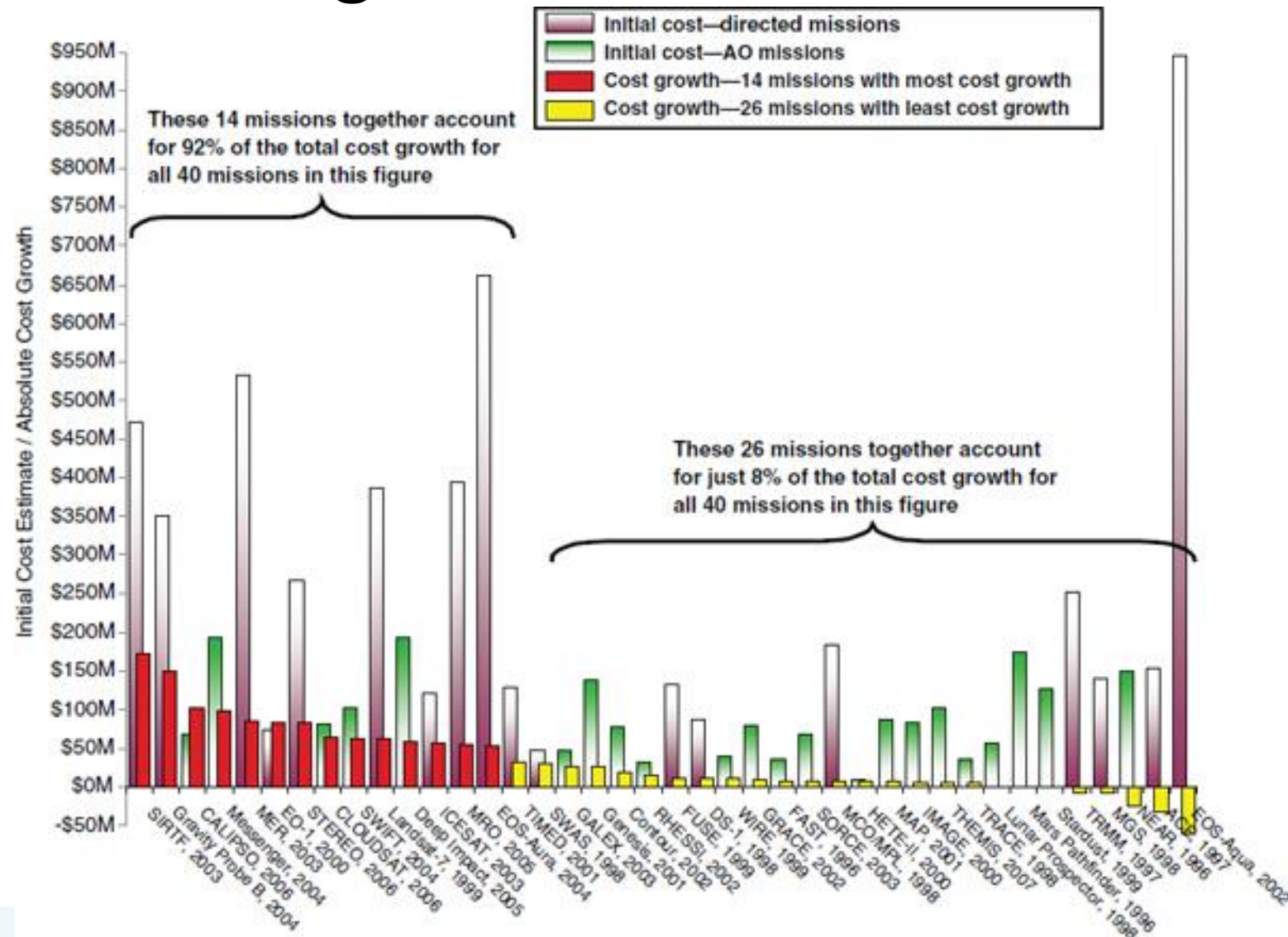
Gentry Lee's Critical Behavioral Characteristics of a Good Systems Engineer



Challenges of current practice ...

- NASA and other formal Systems Engineering processes are very helpful and valuable, **but ...**
 - Assume mostly “clean sheet” design, but many real projects are modifications of previous systems
 - How to do “redesign”, use legacy or COTS components etc...?
 - Assume that system/mission requirements and stakeholder needs are known and stable over time, but in reality they change with new administrations
 - Impact of externalities (e.g. policy) is underrepresented
 - Effect of design iterations and rework on budgets and project outcomes is more important than the linear “waterfall” or “stagegate” process suggests
 - See recent NRC Study on Cost and Schedule Growth in NASA’s Earth and Space Science Missions in which I participated (see next chart).
- Etc...

Ranking of 40 NASA science missions in terms of absolute cost growth in excess of reserves



The Biggest Challenge in Simulation

Q: What is the answer to the question of life, the universe, & everything?

A: 42.

Q: But that answer makes no sense.

A: You do not know the question.

excerpt from *A Hitchhikers Guide to the Universe* by Douglas Addams

Modeling 101

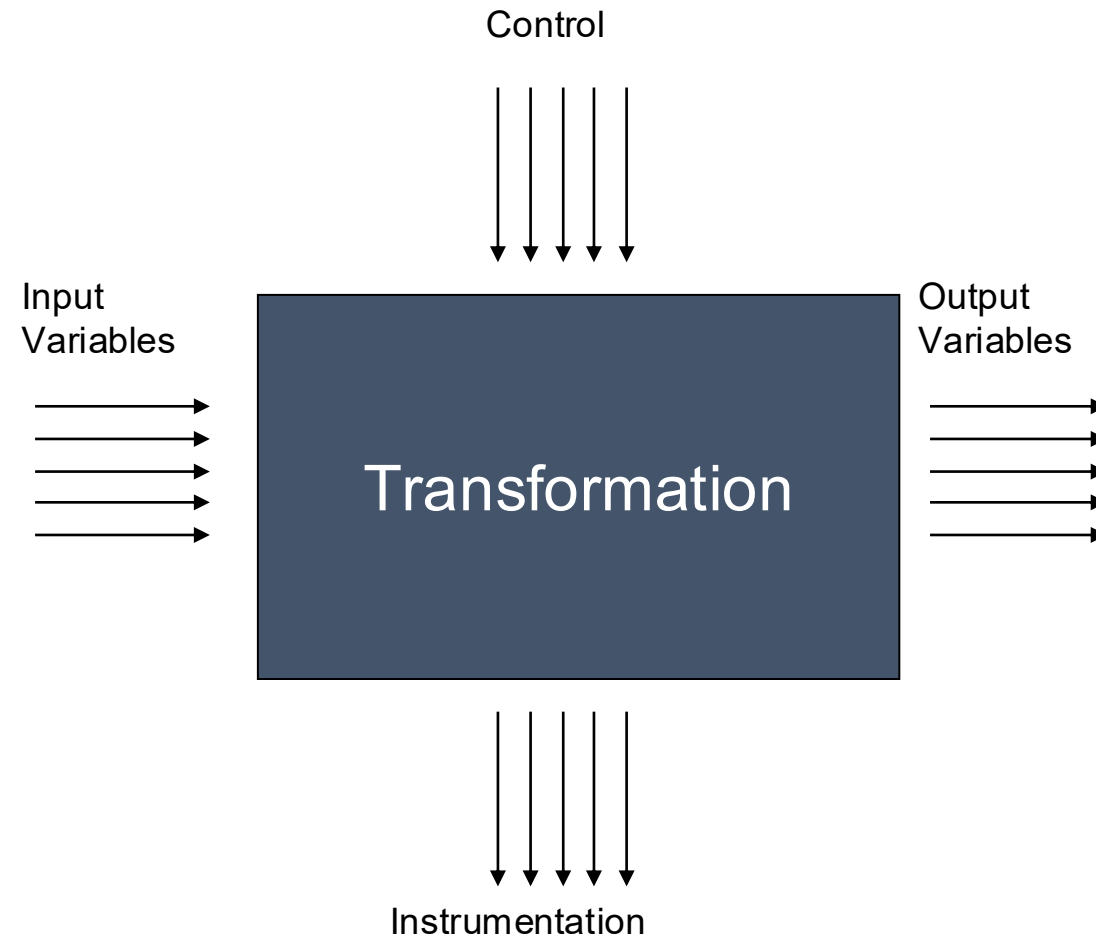
“A mathematical model is an abstract, simplified, mathematical construct related to a part of reality and created for a particular purpose.” (E.A. Bender, *An Introduction to Mathematical Modeling*)

- The first step -- answer the following questions.
 - Why construct a model?
 - What do you wish to conclude?

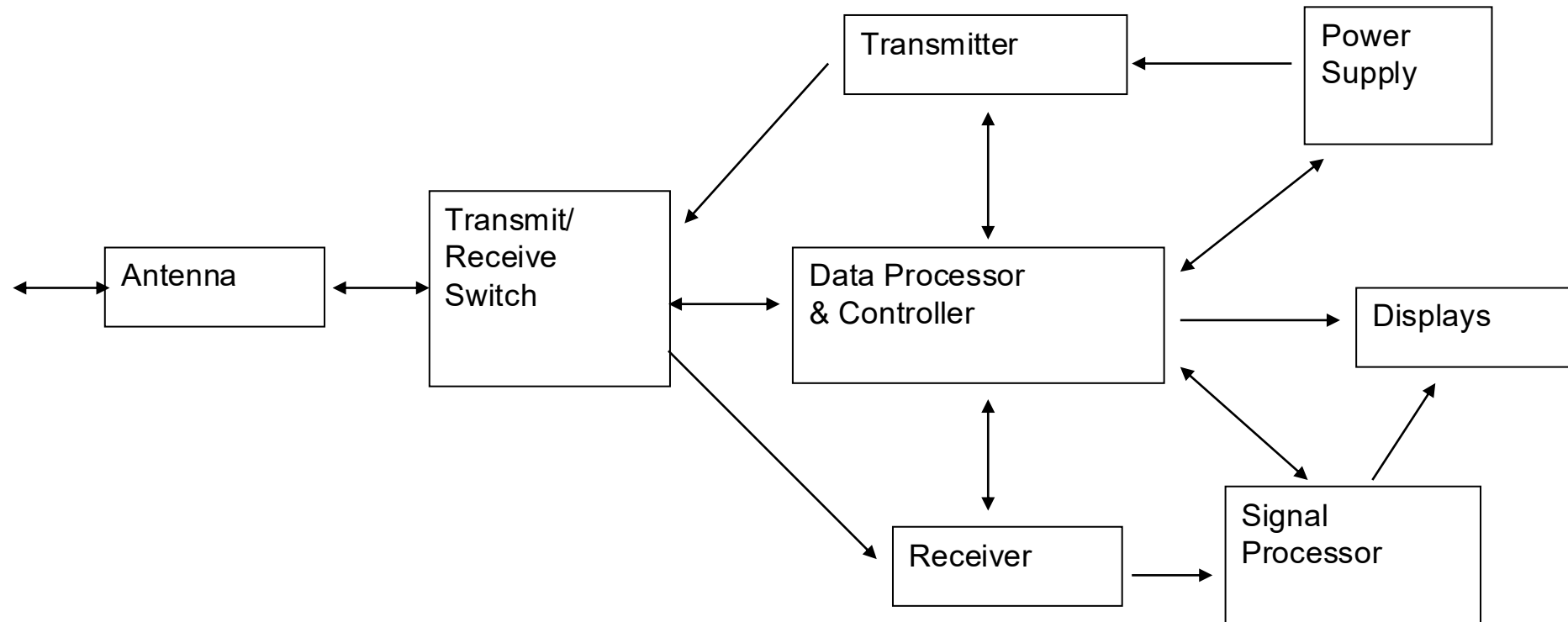
Types of Models

- Deterministic
 - mathematical models
 - lift of an airplane wing
 - thrust of a rocket engine
- Stochastic
 - random discrete event models
 - wind velocities encountered by a flight vehicle during ascent
- Hybrid
 - elements of mathematical & random discrete event models
 - ascent performance of a flight vehicle through the atmosphere

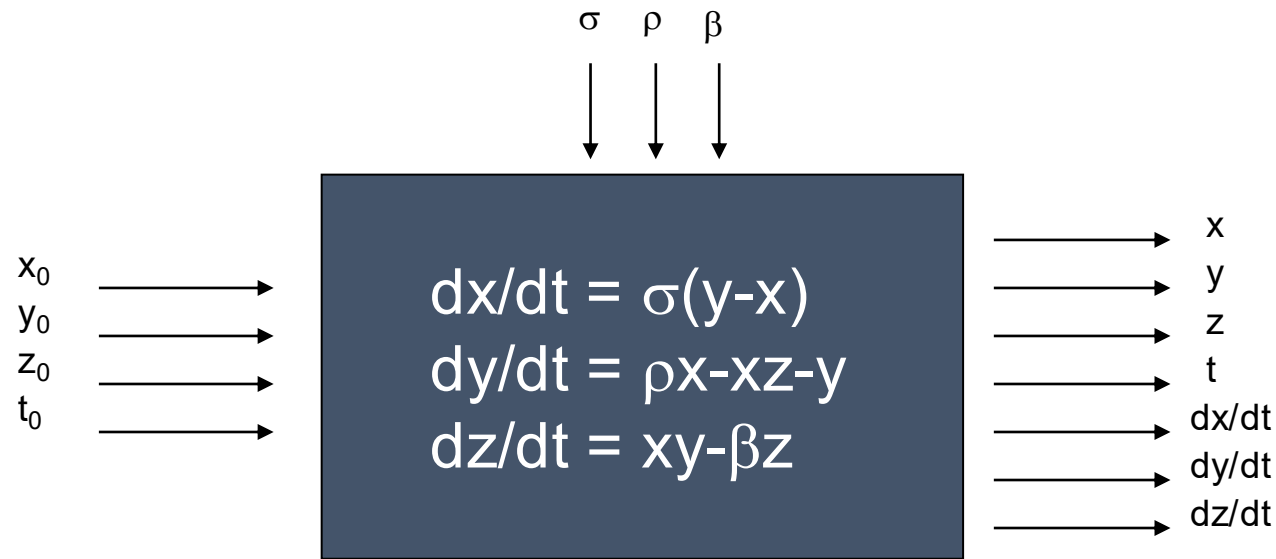
Basic System Component



A Typical System Model -- A Radar System

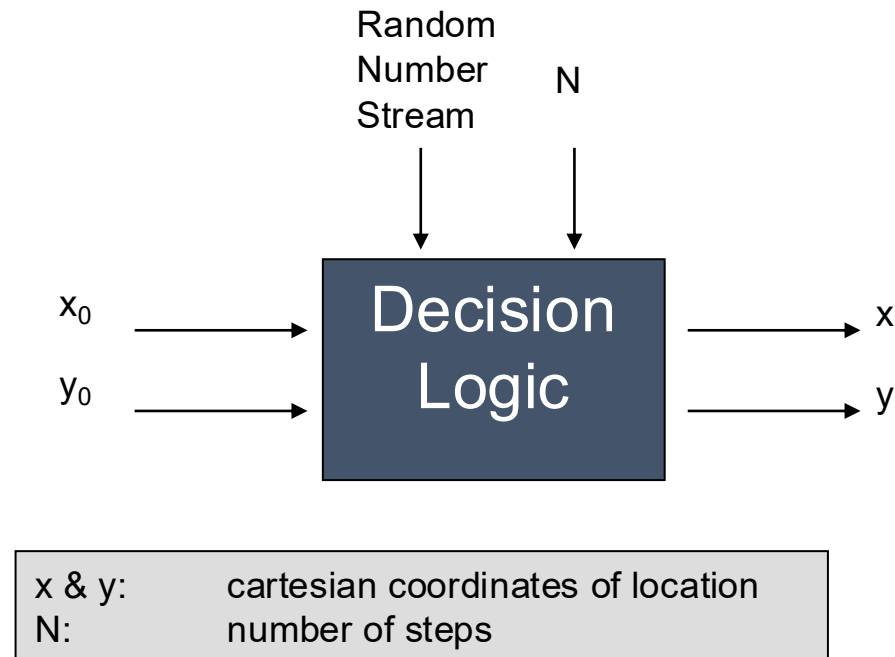


A Deterministic System Model -- Lorenz Model of Atmospheric Dynamics



x, y, z :	cartesian coordinates for surface coordinates & altitude
t :	time
σ :	ratio of viscosity to thermal conductivity (10)
ρ :	nondimensional temperature gradient (28)
β :	geometric factor (8/3)

A Stochastic System Model -- The Random Walk (Staggering Drunk)



A Stochastic System Model -- The Random Walk (Staggering Drunk)

- Model -- defined earlier; an abstract representation of a system
- Simulation -- an imitation of system performance over time to a predefined degree of fidelity
 1. design analyses (model the system & the environment)
 2. breadboards (model the system)
 3. qualification testing (models the environment)
 4. training (simulates the mission)

Learning Objectives

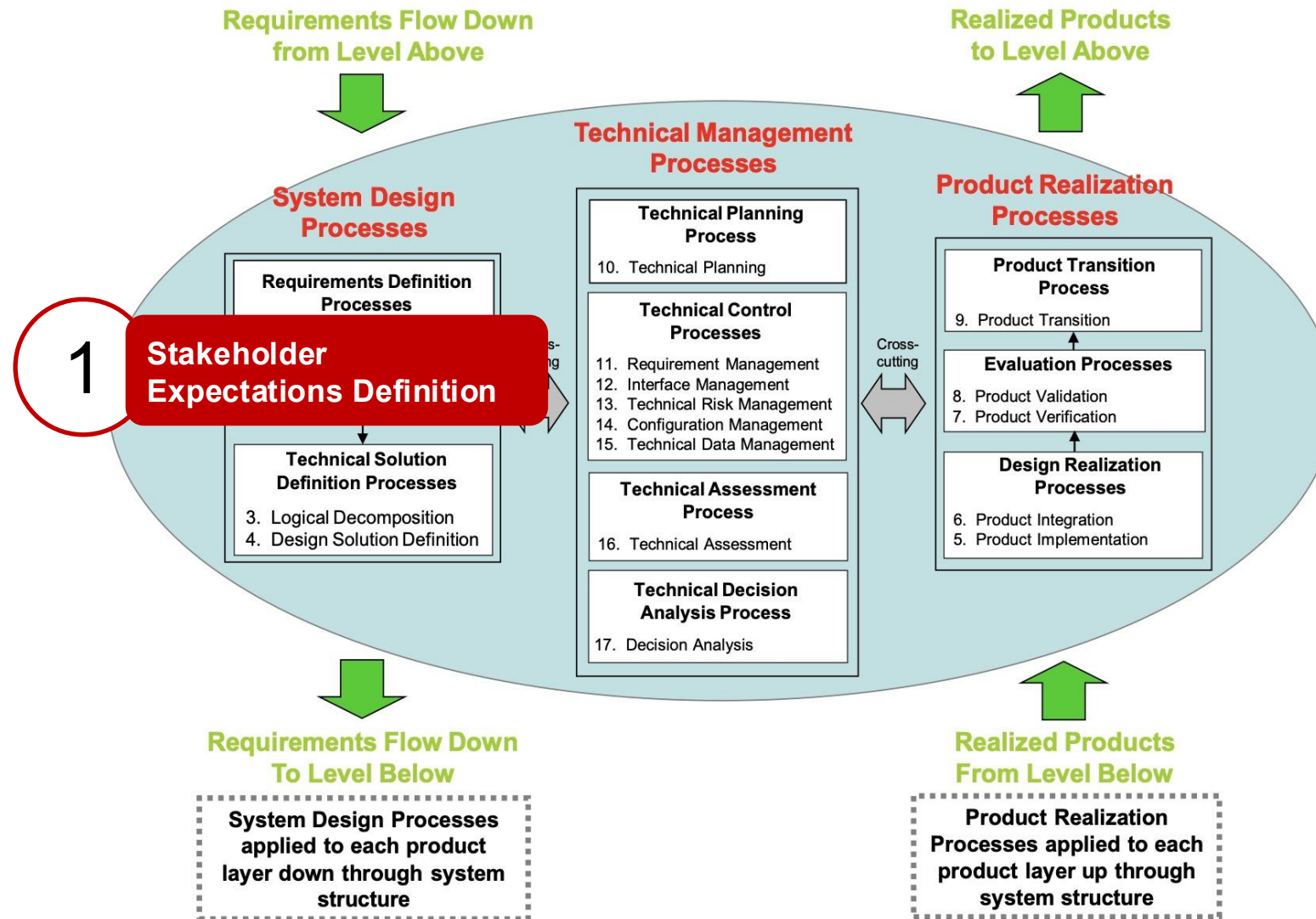
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 - CONOPS
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Concept Question:

- Before rushing to propose or design something we need to spend considerable amount of time **engaging with stakeholders**, Why?
- Pick what you think is the most important reason
 - Because they have the funds
 - Because we need to understand their needs first
 - Who is my competition?
 - Need to understand regulations and laws
 - Other

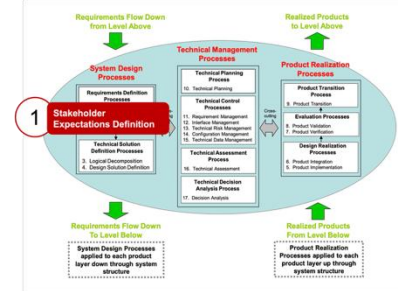
NASA view of stakeholder process

Stakeholder Expectations Definition Process

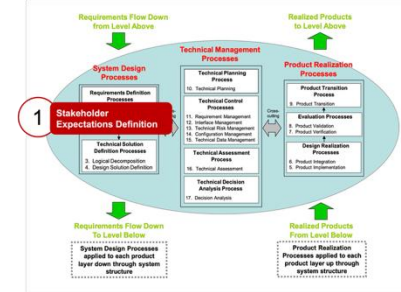


Who is a Stakeholder?

- A group or an individual who is **affected by** or is in some way **accountable** for the outcome of an undertaking
- Stakeholders can be classified as:
 - **Customers** – An organization or individual that has requested a product and will receive the product to be delivered. Examples:
 - An end user of the product
 - The acquiring agent for the end user
 - The requestor of the work product from a technical effort
 - **Other interested parties** who provide broad overarching **constraints** within which the customers' needs must be achieved, or who **have influence** on success of the system. Examples:
 - Those affected by the resulting product
 - Those affected by the manner in which the product is realized or used
 - Those who have a responsibility for providing life-cycle support services (e.g. design, manufacturing, operations, maintenance)



Examples of Stakeholders



Relative to Org	Stakeholder	Typical Expectations
External	Customer	Expected level of product quality, delivered on time, affordable, life cycle support & services
	Subcontractors/vendors	Well defined requirements
	Local, State, National Public	Products must not contaminate the environment
Internal	Org Management	Internal Commitments met (cost, schedule), good status provided, compliance with org policies, directives and procedures
	PM	Expected technical work products delivered on time and can be used for decision making
	Technical Team members	clear tasks, job security, rewards, teamwork
	Functional Organizations (e.g., test)	Test support products available, clear test requirements, recognition for project help

The diagram illustrates the Systems Engineering process as a continuous cycle between four main phases, each with specific sub-processes:

- System Design Processes**
 - 1. Requirements Definition Processes
 - 2. Technical Solution Definition Processes
 - 3. Logical Decomposition
 - 4. Design Solution Definition
- Technical Management Processes**
 - Technical Planning Processes
 - 1. Technical Planning
 - Technical Control Processes
 - 1. System Performance Monitoring
 - 2. System Configuration Management
 - 3. Configuration Management
 - 4. Change Management
 - 5. Configuration Management
 - Technical Assessment
 - 1. Technical Assessment
 - Technical Management Analysis Processes
 - 1. Decision Analysis
- Product Realization Processes**
 - Product Transition Processes
 - 1. Product Transition
 - 2. Evaluation Processes
 - 3. Production Processes
 - Deployment Processes
 - 1. Deployment Processes
 - Product Implementation Processes
 - 1. Product Implementation
- Stakeholder Expectations Definition** (Highlighted with a red arrow labeled '1')

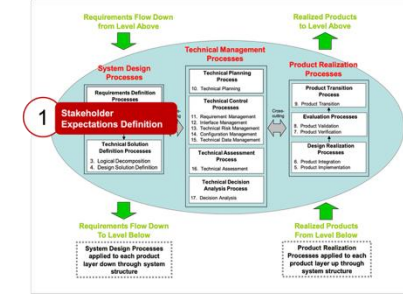
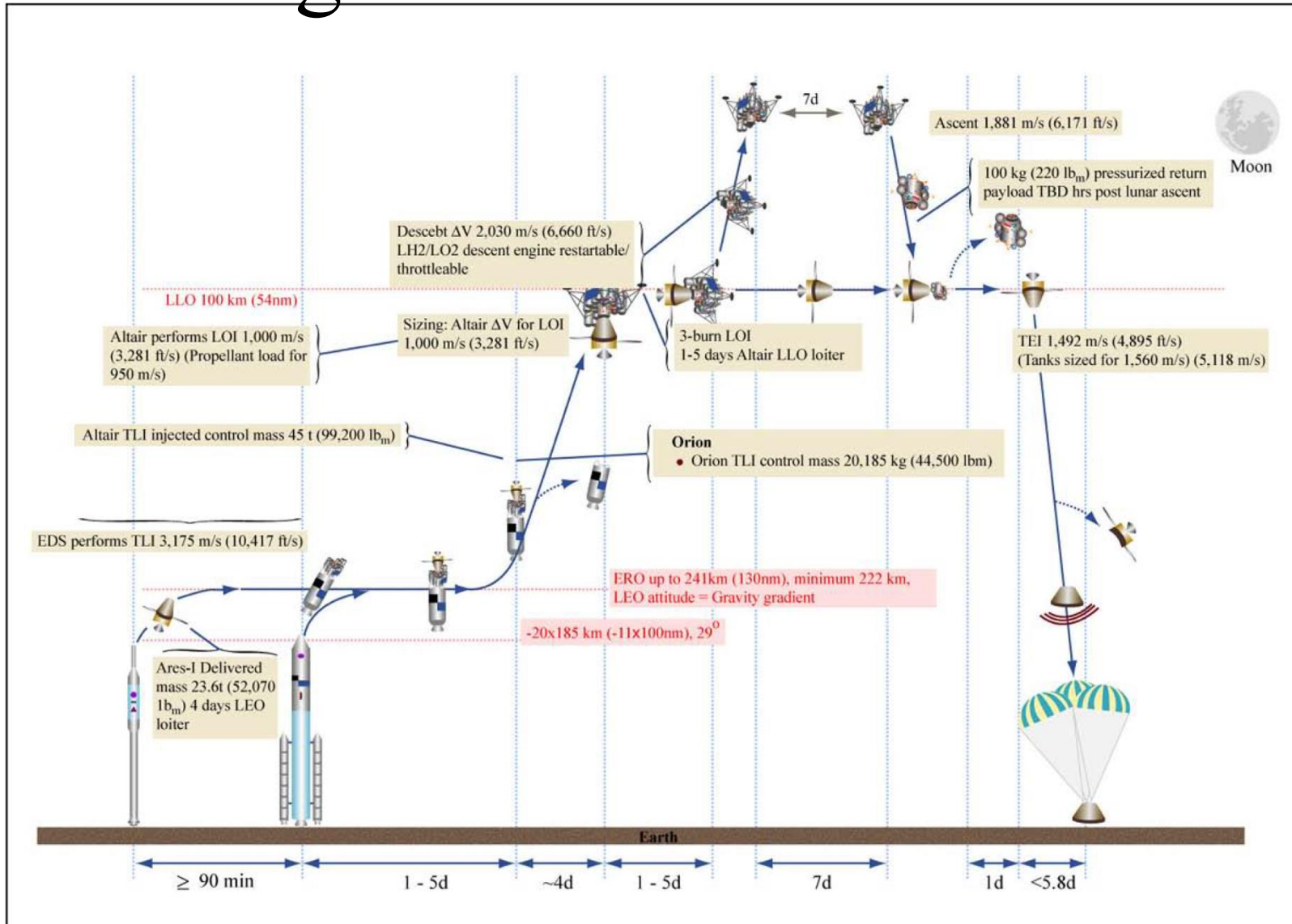
The flow of the process is indicated by green arrows:

- Requirements Flow Down from Level Above** (into System Design Processes)
- Realized Products to Level Above** (from Product Realization Processes)
- Realized Products from Level Below** (into Product Realization Processes)
- Requirements Flow Down to Level Below** (from System Design Processes)

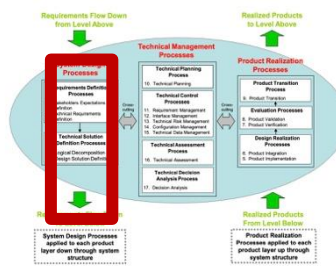
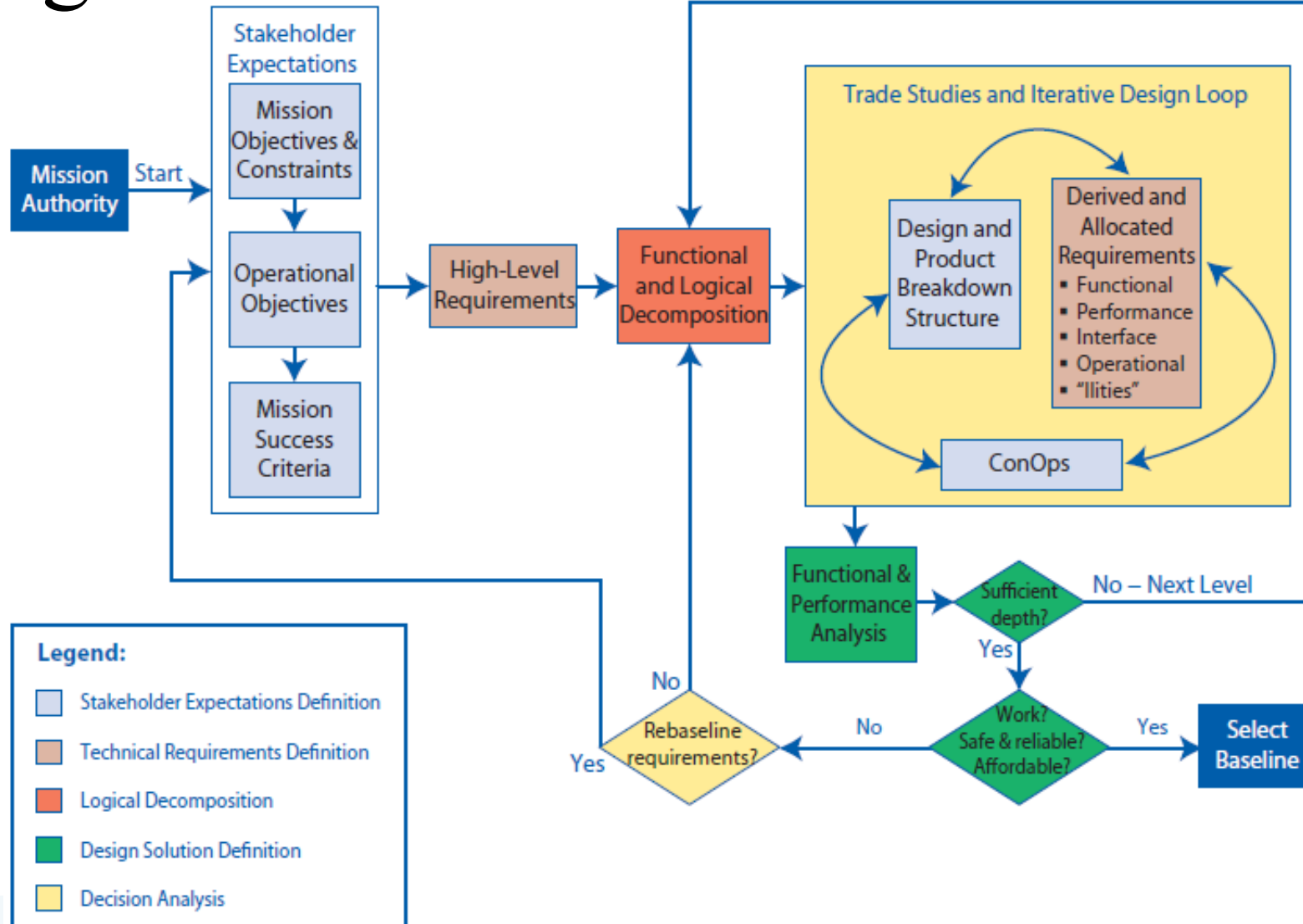
Additional notes:

- System Design Processes** applied to each product layer down through system structure.
- Product Realization Processes** applied to each product layer through system structures.

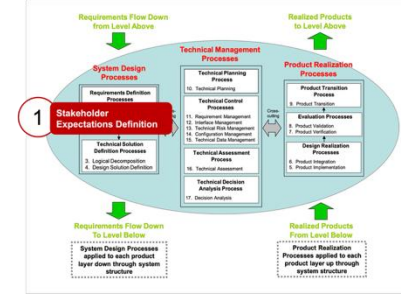
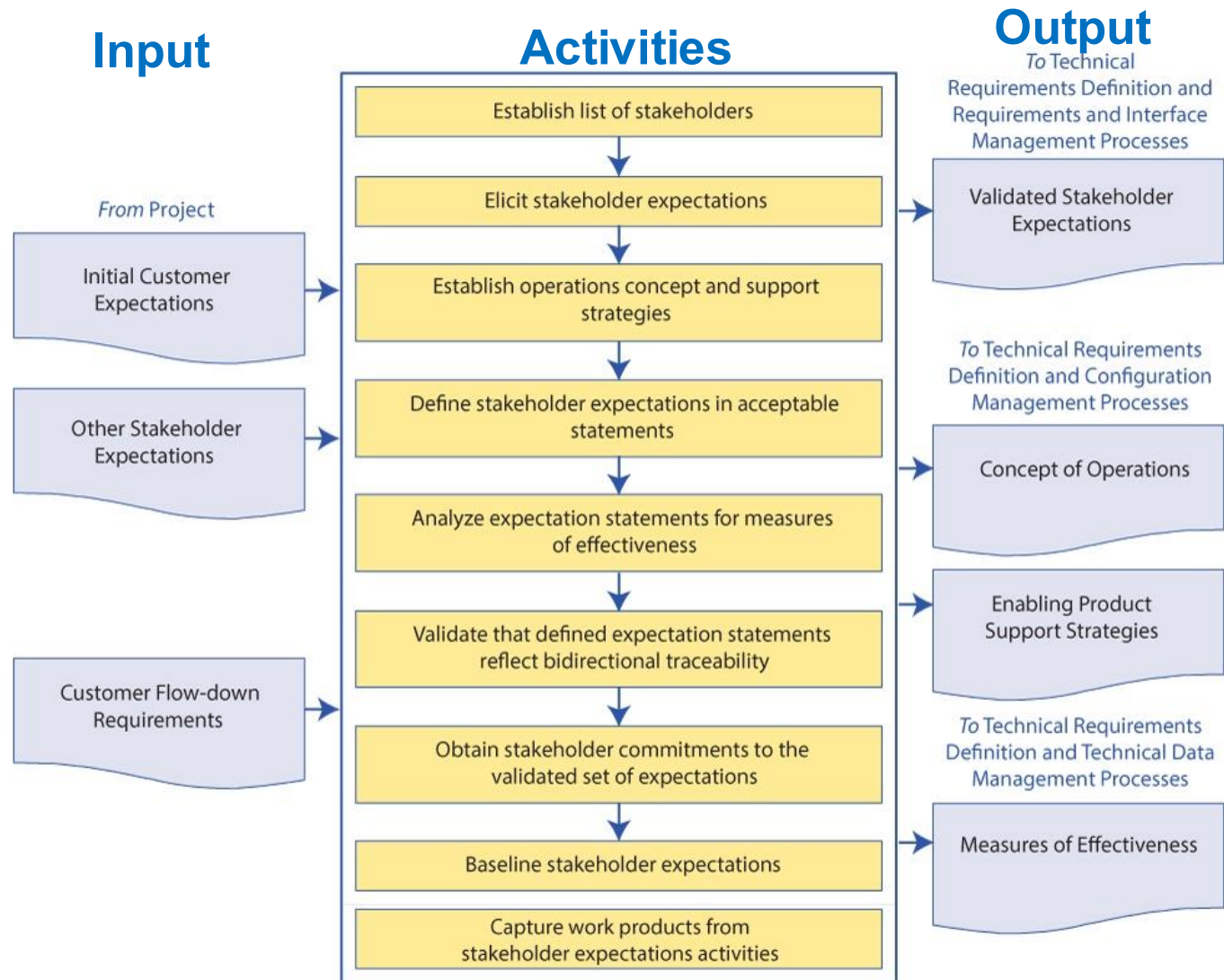
Example CONOPS: Lunar Design Reference Mission



Relationships among the upstream System Design Processes



Stakeholder Expectations Definition - Best Practice Process Flow Diagram



What are the Benefits of the Stakeholder Expectations Process?

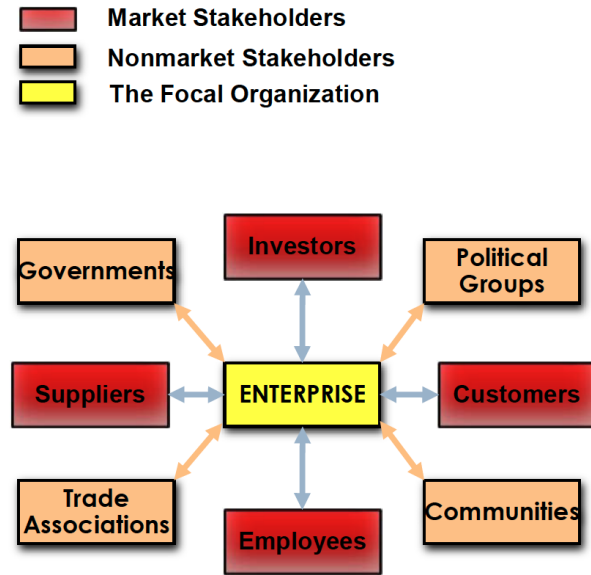
- Build a system that meets customers' **expectations**
 - Operators requirements
 - Support from Congress and Public
- Build a system that can be **tested, operated** and **maintained**
- Ensure Stakeholder **commitments** are obtained and realized.

Stakeholder Value Network (SVN) Modeling

- Most stakeholder models only focus on a single focal organization and ignore the indirect relationships amongst other stakeholders. This can lead to project failures if not recognized. Stakeholder Value Network (SVN) models attempt to capture these 2nd order effects and value loops.

Reading [1d] summarizes this approach: Feng W., Crawley E.F., de Weck O.L., Keller R., Robinson R., “Dependency Structure Matrix Modeling for Stakeholder Value Networks”, 12th International Dependency and Structure Modeling Conference, DSM’10, Cambridge, UK, 22-23 July 2010

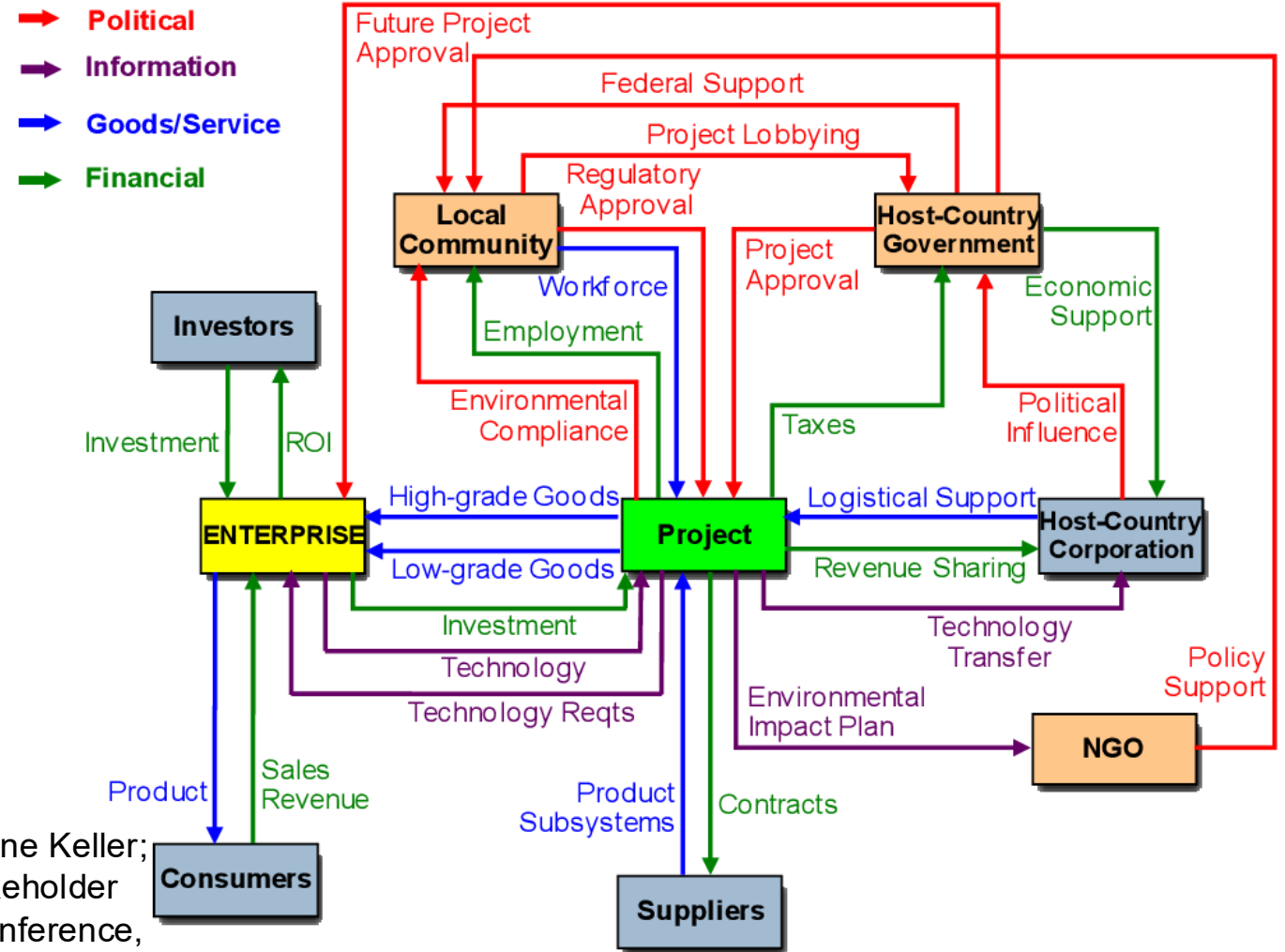
Introduction: “Hub-and-Spoke” Model and Value Network



“Hub-and-Spoke” Stakeholder Model
(Adapted from Donaldson and Preston, 1995)

Figure 3 in Wen Feng; Edward F. Crawley; Olivier de Weck; Rene Keller; Bob Robinson. “Dependency structure matrix modelling for stakeholder value networks”. Proceedings of the 12th International DSM Conference, Cambridge, UK, 22.–23.07. 3–16. DSM 2010.

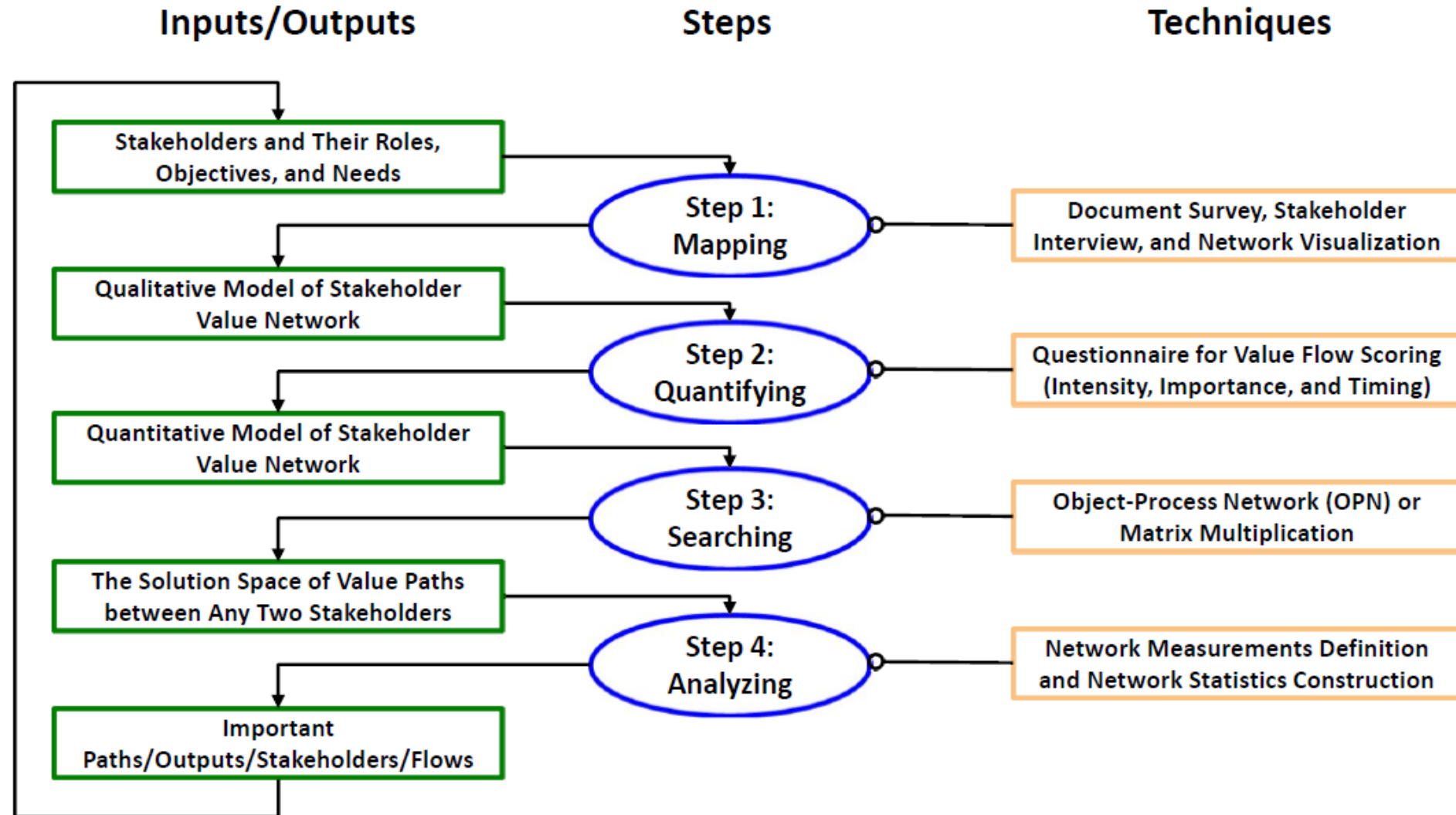
Stakeholder Value Network (Feng, Cameron, and Crawley, 2008)



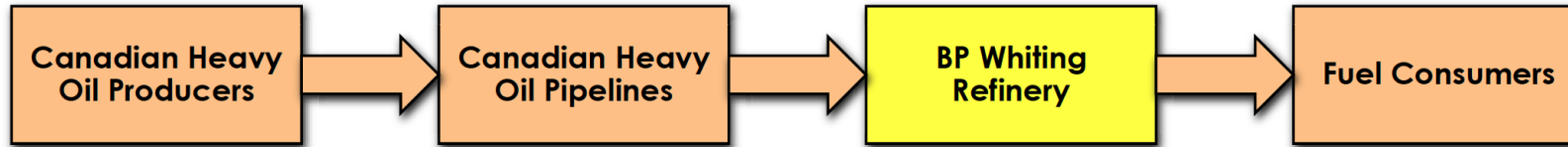
Motivation

- **Indirect Relationships:**
 - Understand the impact of both direct and indirect relationships between stakeholders on the success of large projects.
- **Strategies with Reduced Complexity:**
 - Apply such an understanding to inform decisions on stakeholder management strategies in a positive way and with reduced complexity.
- **Communication Platform:**
 - Build a common platform for engineering, external affairs, and management within a project to consistently communicate important information about stakeholders.

SVN Methodology



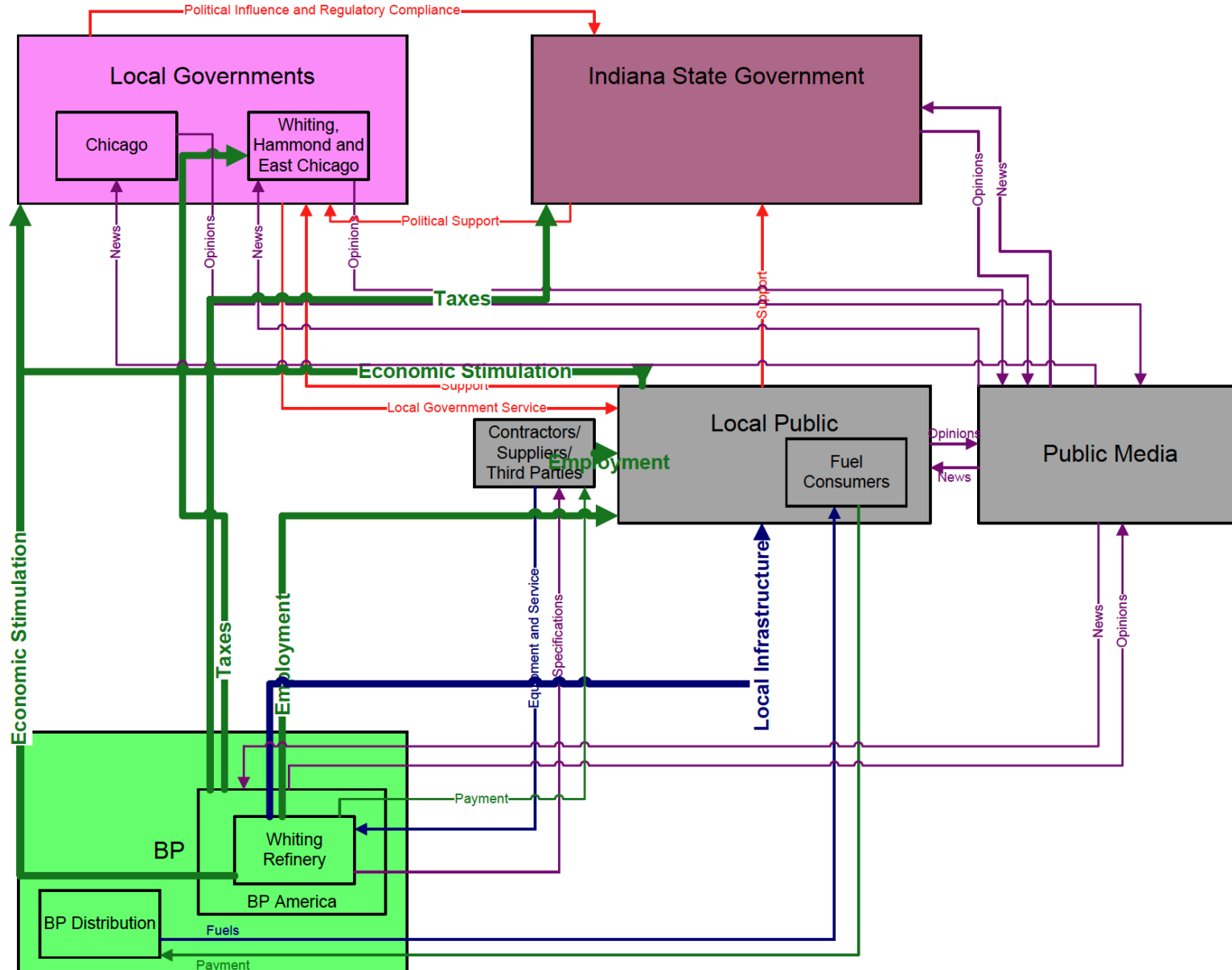
A Case Study from the Oil & Gas Industry



BP Whiting Refinery Modernization Project

- **Whiting Refinery:** located in northwest Indiana, with more than 100-year history;
- **Modernization Project:** 2007-2012, \$3.8 billion, + 1.7-million-gallon gasoline/day;
- **Stakeholder Support:** new water permits for increased discharge of ammonia and suspended solids, issued by Indiana Department of Environmental Management (IDEM) and EPA;
- **Stakeholder Opposition:** an **unanticipated** firestorm of protest from Public Media (Chicago Tribune, Chicago Sun-Times, etc.) and Local Public (in Illinois), concerning water permitting;
- **BP finally promised to keep the lower discharge limits**
- **Managers' initial Mental Model for Stakeholder Importance:** Indiana State Government (IDEM) and U.S. Federal Government (EPA) are the most important stakeholders.

Issue 1: Local Economic Stimulation - *Issue Map*



Markov processes

“The future is independent of the past given the present”

$$\textit{New State} = f(\textit{old state}, \textit{noise})$$

Definition: A state S_t is Markov if and only if

$$P[S_{t+1} | S_t] = P[S_{t+1} | S_1, \dots, S_t]$$

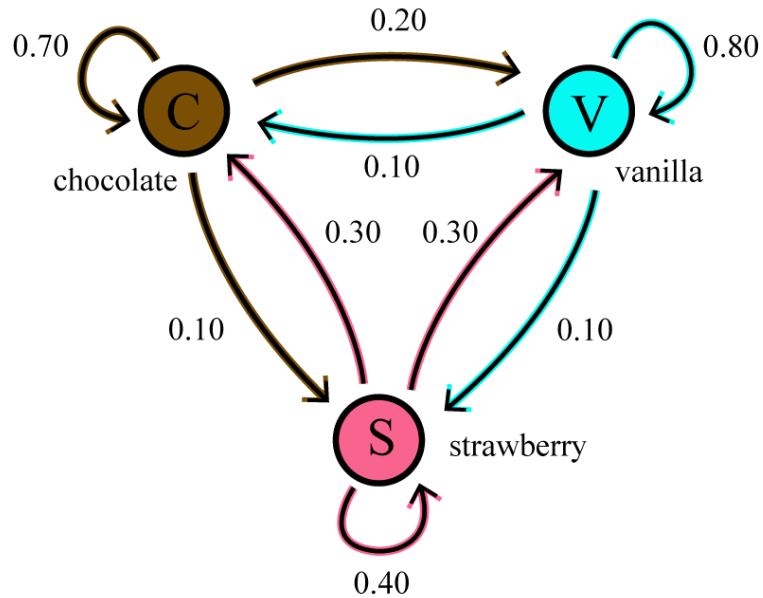
- The state captures all relevant information from the history
- Once the state is known, the history may be thrown away
- i.e. The state is a sufficient statistic of the future



Galton board or bean machine



Markov processes



Chocolate	Strawberry	Vanilla	
70%	30%	10%	Chocolate
10%	40%	10%	Strawberry
20%	30%	80%	Vanilla

Calcworkshop.com

$$P = \begin{bmatrix} 0.7 & 0.3 & 0.1 \\ 0.1 & 0.4 & 0.1 \\ 0.2 & 0.3 & 0.8 \end{bmatrix} \longrightarrow P' = \begin{bmatrix} 0.32 \\ 0.14 \\ 0.54 \end{bmatrix}$$

Markov processes – example

Queueing theory

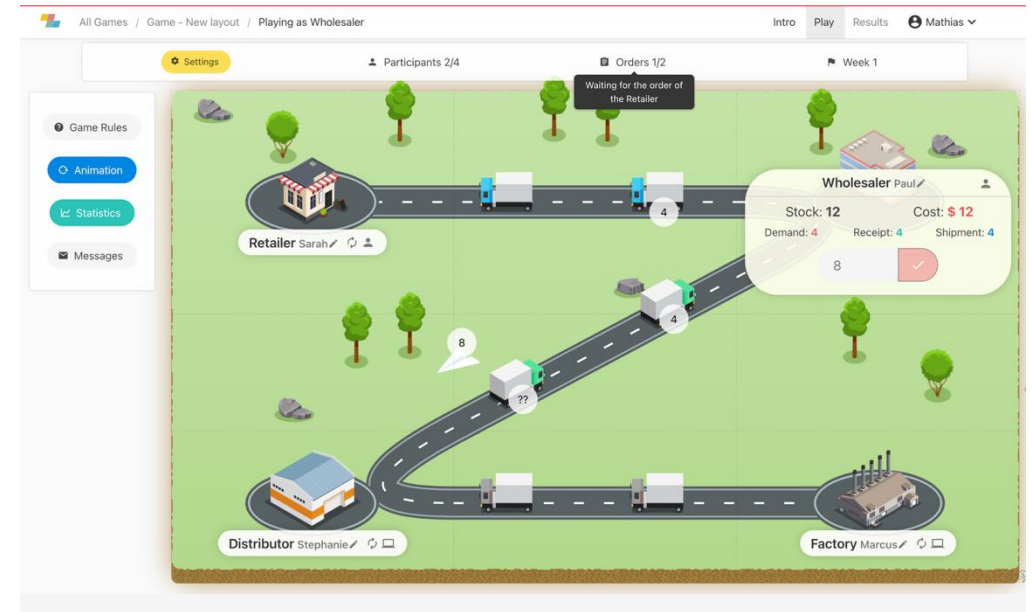


Queueing theory

Queueing theory

Inventory theory

- Economic Order Quantity (EOQ)
 - EOQ's Assumptions:
 - Demand rate for the product is known and constant over time
 - Time between placing an order and receiving it is fixed and predictable
 - Entire quantity is received at once, not gradually over time
 - Inventory is managed so that shortages do not occur; new orders arrive just before inventory reaches zero.
 - Cost of placing an order is constant regardless of order size
 - Single product with independent demand
- The Beer Distribution Game



Inventory theory

Inventory theory - Example

A manufacturing company produces electric motors and uses precision ball bearings as a key component in its assembly process. The plant expects to use approximately 24,000 bearings per year in production. The cost of placing an order with the supplier is \$150 per order, and the annual holding cost is \$8 per bearing. The manufacturing plant operates 300 days per year.

- a. What is the Economic Order Quantity (EOQ) for the bearings?
- b. How many orders per year should the plant place?
- c. What is the expected time between orders (in days)?
- d. What is the total annual inventory cost if the EOQ quantity is ordered?

Systematic Literature Review (SLR)

- **What is a Literature Review?**

- a collection of all the **scholarly** writing on a topic
- an **integrative** summary of published research on a specific topic
- Seeks to **synthesize** what is already known about the topic, and sometimes, explicitly state what is not known, or not well understood.
- map the research **terrain** or **scope**
- systemize **relationships** between concepts
- identify **gaps** in the literature

- **What does the literature show us?**
- **Connect your idea to the literature.**
- **Survey the literature on the topic.**
- but what IS 'the literature'?

Systematic Literature Review (SLR)



- Books



- Scholarly articles



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- Conventions
- Annual Meetings
- Conference Proceedings

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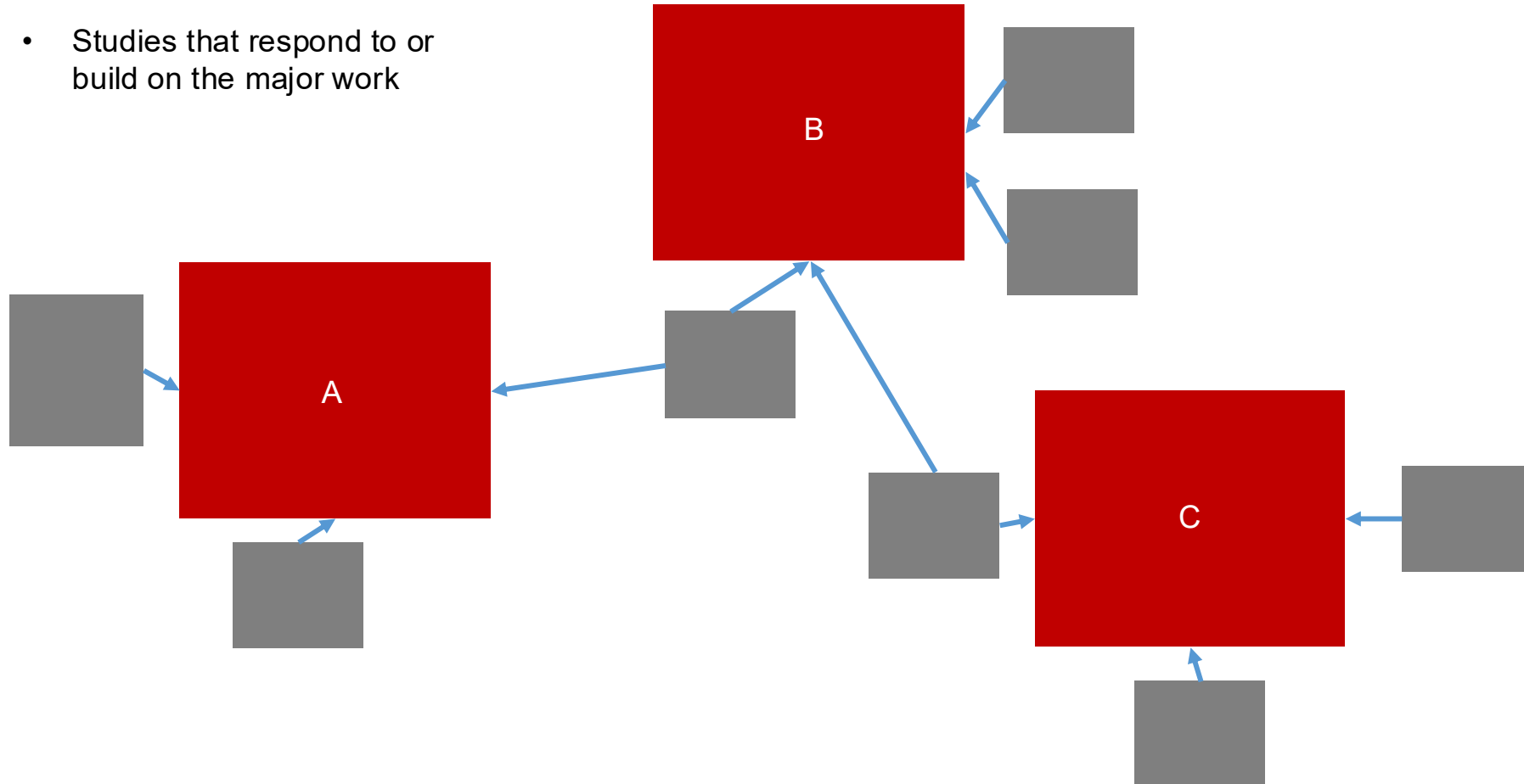
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UNIVERSITY OF ALABAMA IN HUNTSVILLE

You are searching 11 databases

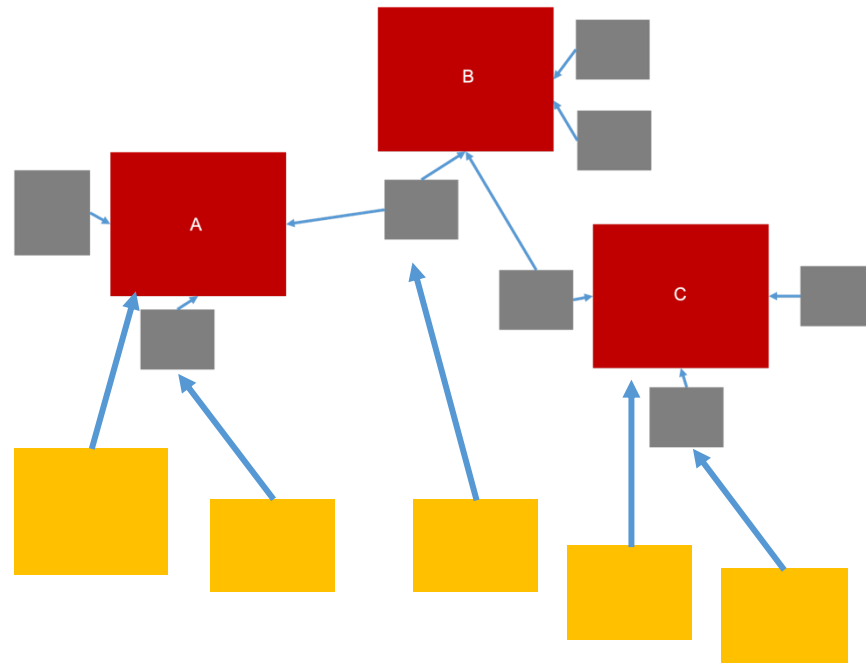
- Dissertations
- Thesis

Systematic Literature Review (SLR)

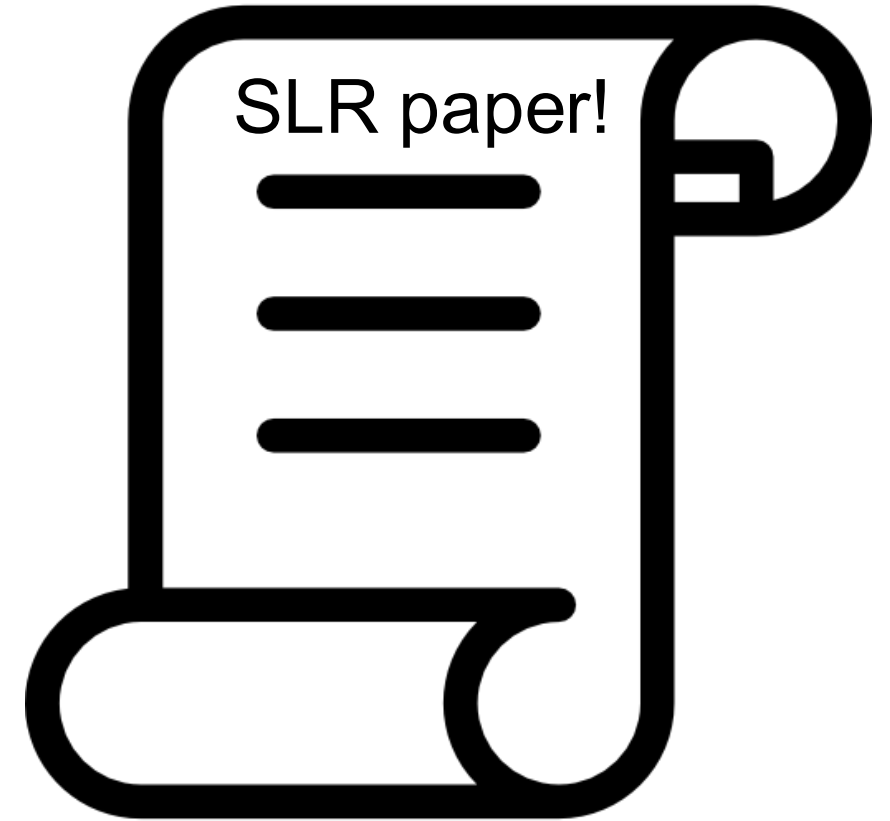
- Major Work
- Studies that respond to or build on the major work



Systematic Literature Review (SLR)

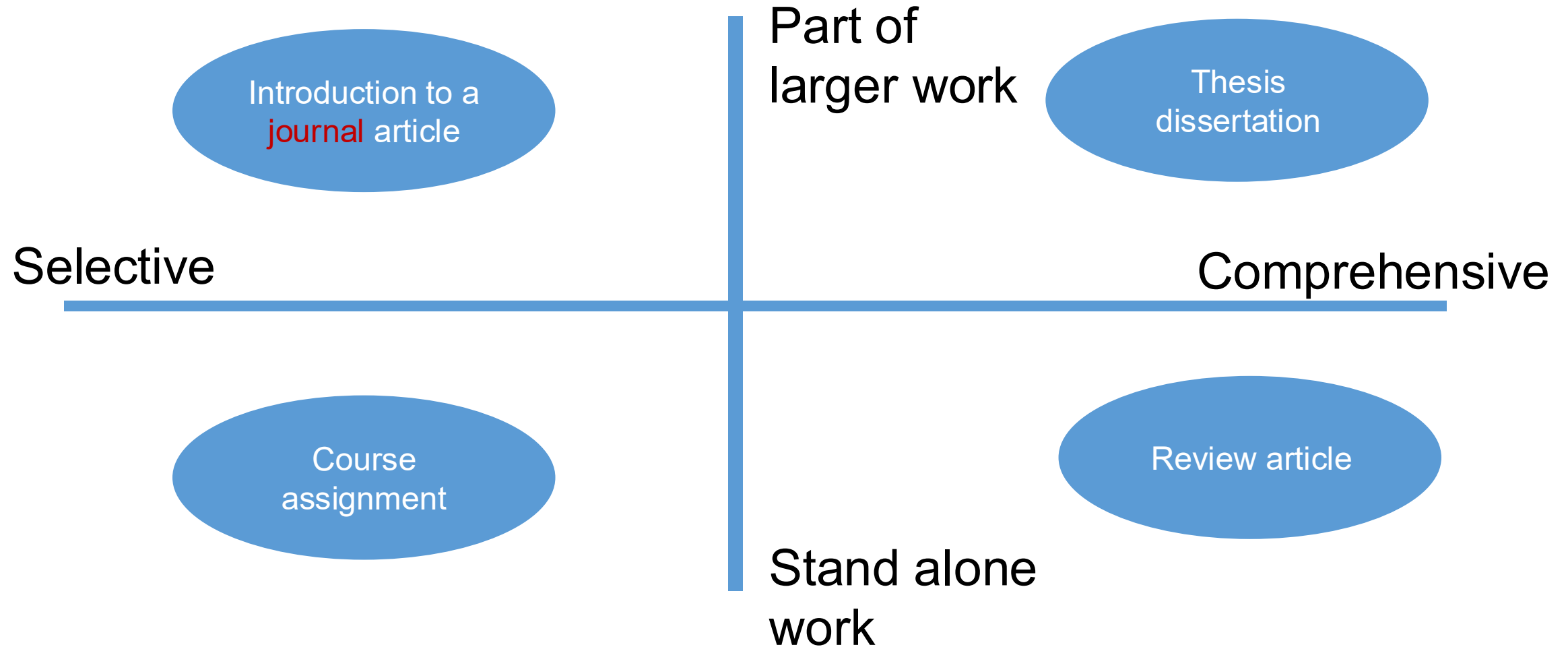


My ideas!



A literature review is not simply a summary of existing studies. Instead, you should use the literature to develop and support your own argument. - Julie P. Martin

Types of Literature Reviews



EDITORIAL



Rethink literature review of design research in an age of AI – from ‘secretary work’ to scholarly synthesis of insight, frameworks, and foresight

Roger Jiao

Georgia Institute of Technology, Atlanta, GA, USA

ABSTRACT

Literature reviews in design research are meant to consolidate knowledge, clarify methodological patterns, and guide future inquiry through synthesis rather than compilation. Yet many published reviews remain largely descriptive – dominated by inventories, procedural narration, and bibliometric counting – and consequently fail to provide a coherent technical framework or actionable insight. AI-generated text further exacerbates this issue by automating surface-level summarisation while introducing risks of fabricated or erroneous citations when outputs are not rigorously verified. This perspective argues that review scholarship must shift from AI-like ‘secretary work’ to genuine knowledge construction, emphasising framework-first synthesis, defensible cross-study comparison, explicit validity regimes and failure modes, benchmark-aligned evaluation, and auditable provenance practices. Building on systematic reporting principles (e.g., PRISMA) while treating reporting as a floor rather than the endpoint, the article outlines a synthesis-driven, auditable review pipeline and calls for coordinated action among authors, reviewers, editors, and the broader community to elevate standards, strengthen integrity, and restore literature reviews as central, high-impact contributions to design research.

ARTICLE HISTORY

Received 31 January 2026
Accepted 31 January 2026

KEYWORDS

Design research
methodology; literature
review; research context;
engineering design

1. Obvious flaws in current review practice, now magnified by AI

Literature reviews in engineering design are intended to consolidate knowledge, clarify methodological patterns, and guide future research. Yet many published reviews offer limited analytical value: they document search procedures, report publication statistics, and summarise papers, but seldom develop integrative frameworks or derive actionable insights for the engineering design community. At a moment when AI systems can automate much of this procedural work, the discipline must elevate expectations for what constitutes a meaningful review. This article highlights recurring shortcomings and proposes a more rigorous, synthesis-driven standard for design review scholarship.

Box 7. Nature’s abstract-writing template

nature

How to construct a *Nature* summary paragraph

Annotated example taken from *Nature* 435, 114–118 (5 May 2005).

One or two sentences providing a **basic introduction** to the field, comprehensible to a scientist in any discipline.

Two to three sentences of **more detailed background**, comprehensible to scientists related disciplines.

One sentence clearly stating the **general problem** being addressed by this particular study.

One sentence summarising the **main result** (with the words ‘**here we show**’ or their equivalent).

Two or three sentences explaining what the **main result** reveals in direct comparison to what was thought to be the case previously, or how the main result adds to previous knowledge.

One or two sentences to put the results into a **more general context**.

Two or three sentences to provide a **broader perspective**, readily comprehensible to a scientist in any discipline, may be included in the first paragraph if the editor considers that the accessibility of the paper is significantly enhanced by their inclusion. Under these circumstances, the length of the paragraph can be up to 300 words. (The above example is 190 words without the final section, and 250 words with it).

During cell division, mitotic spindles are assembled by microtubule-based motor proteins^{1,2}. The bipolar organization of spindles is essential for proper segregation of chromosomes, and requires plus-end-directed homotetrameric motor proteins of the widely conserved kinesin-5 (BimC) family³. Hypotheses for bipolar spindle formation include the ‘push–pull mitotic muscle’ model, in which kinesin-5 and opposing motor proteins act between overlapping microtubules^{4,5}. However, the precise roles of kinesin-5 during this process are unknown. Here we show that the vertebrate kinesin-5 Eg5 drives the sliding of microtubules depending on their relative orientation. We found in controlled *in vitro* assays that Eg5 has the remarkable capability of simultaneously moving at $\sim 20 \text{ nm s}^{-1}$ towards the plus-ends of each of the two microtubules it crosslinks. For anti-parallel microtubules, this results in relative sliding at $\sim 40 \text{ nm s}^{-1}$, comparable to spindle pole separation rates *in vivo*⁶. Furthermore, we found that Eg5 can tether microtubule plus-ends, suggesting an additional microtubule-binding mode for Eg5. Our results demonstrate how members of the kinesin-5 family are likely to function in mitosis, pushing apart interpolar microtubules as well as recruiting microtubules into bundles that are subsequently polarized by relative sliding. We anticipate our assay to be a starting point for more sophisticated *in vitro* models of mitotic spindles. For example, the individual and combined action of multiple mitotic motors could be tested, including minus-end-directed motors opposing Eg5 motility. Furthermore, Eg5 inhibition is a major target of anti-cancer drug development, and a well-defined and quantitative assay for motor function will be relevant for such developments.

Nature’s abstract-writing template, <https://www.nature.com/nature/for-authors/formatting-guide>. © 2021 Springer Nature Limited. All rights reserved. Permission for further reuse must be obtained from the relevant holder of the exclusive rights.

(Grant and Booth 2009)



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Structural Change and Economic Dynamics
17 (2006) 127–147

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The shift to services employment: A review of the literature

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Received December 2003; received in revised form January 2005; accepted April 2005
Available online 28 November 2005

Abstract

This paper provides an overview of literature on the employment shift towards services. It follows the three dimensions of structural change: inter-industry productivity differences, the inter-industry division of labor (outsourcing), and shifts in final demand. It concludes that the third dimension gained importance over the last decades although differential productivity growth continued to contribute to the rise of service employment as well. Outsourcing of service tasks from manufacturing industries increased but cannot explain much of the rise in service-industry employment over time. The shift to services is not just a price effect nor is it mainly the effect of the outsourcing of service activities from manufacturing industries. The shift to services is real.

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JEL classification: E2; J2; L0; O1; O4

Keywords: Consumption, saving, production, employment, and investment; Time allocation, work behavior, employment determination and creation; Industrial organization: general; Economic development; Economic growth and aggregate productivity

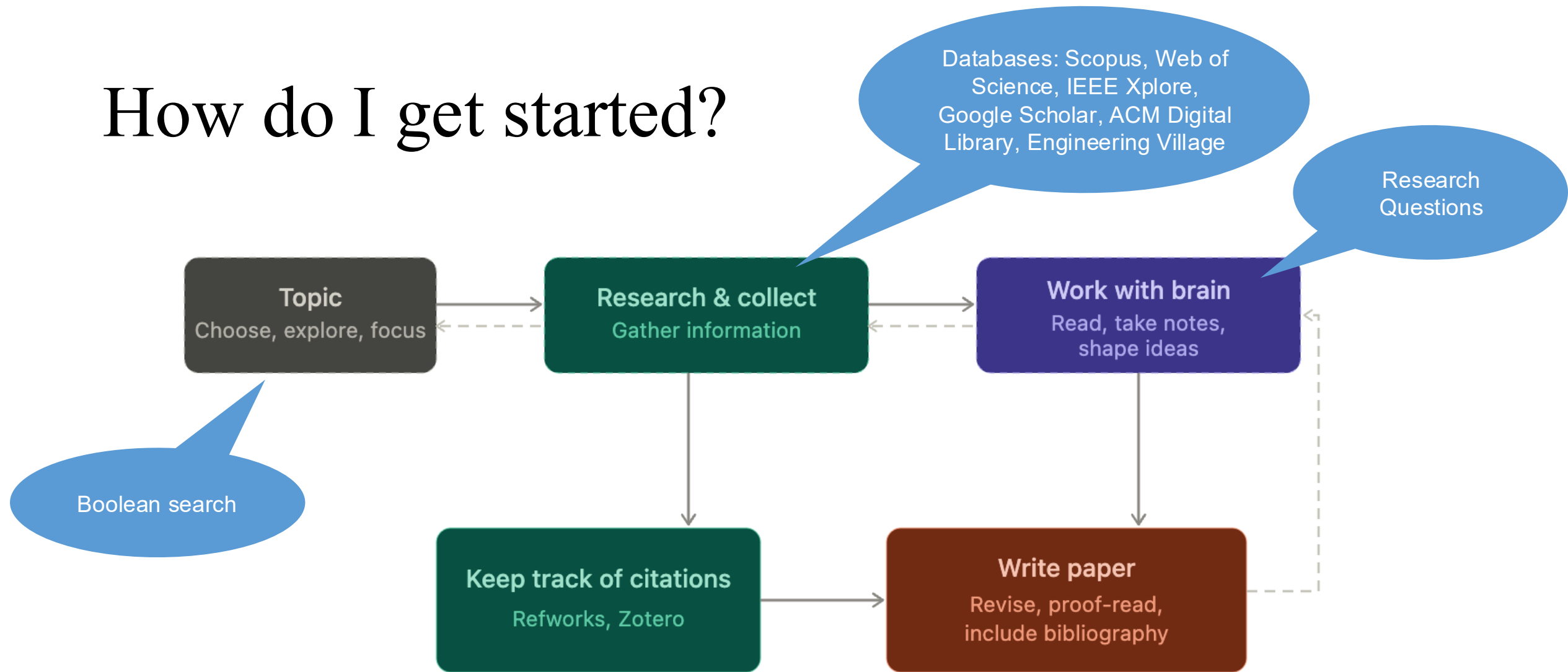
Stand alone article

Not all review article will
say literature reviews in
the title

Reading Assignment:

<https://chemrxiv.org/doi/pdf/10.26434/chemrxiv.15005987/v1>

How do I get started?



How do I review the literature I find?

- Collect & read relevant literature
- Provide an overview of significant literature
- Highlight key concepts & papers



Describe

Organize

Summarize

Analyze

Compare & contrast

Critically evaluate

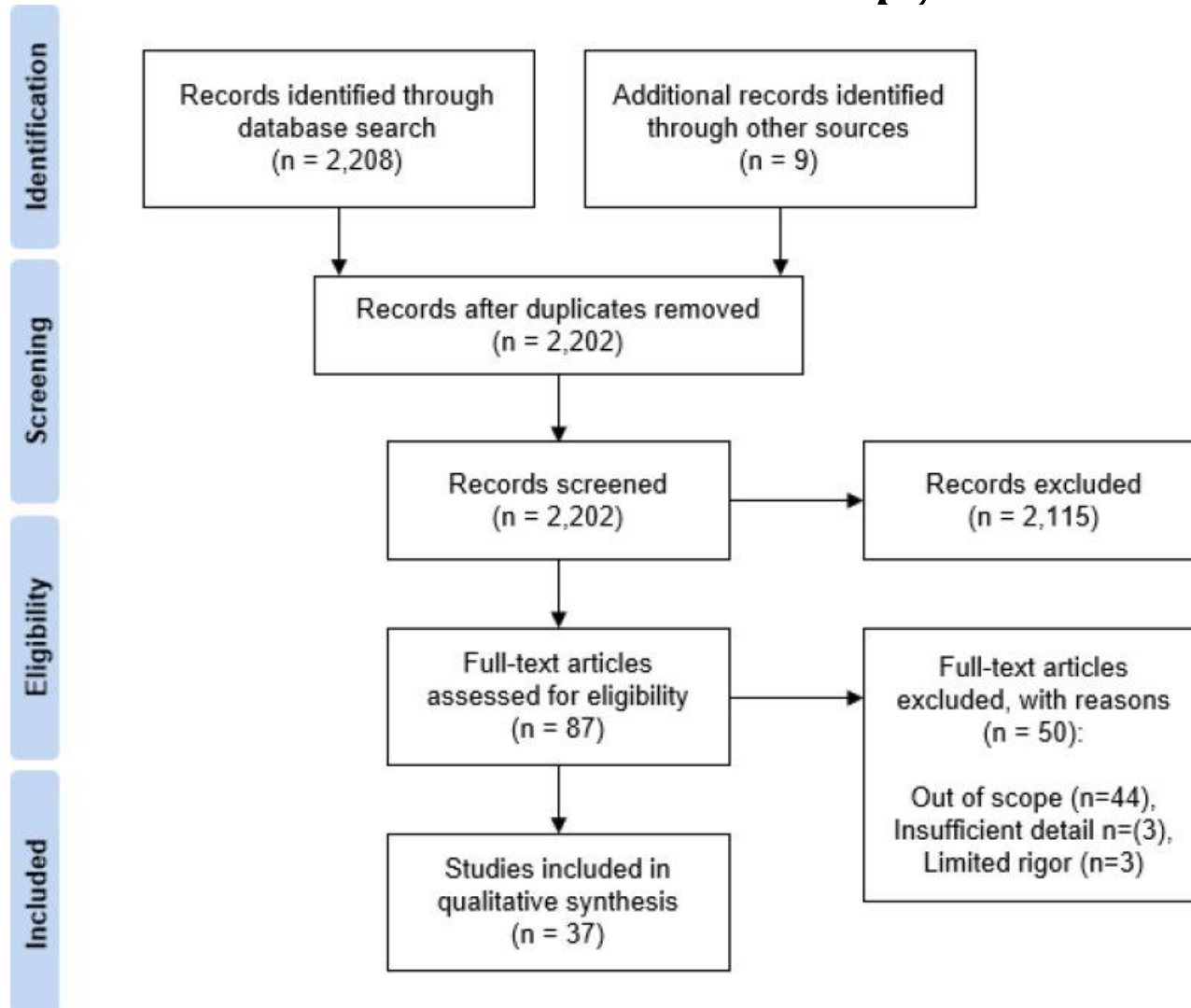
What's involved in Reviewing?

- Discover relationship between sources
- Identify major themes & concepts
- Identify critical gaps & disagreements

Don't turn the literature review into a laundry list of study summaries.



PRISMA Flow Diagram



<https://www.prisma-statement.org/prisma-2020-flow-diagram>

References:

My Library

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The making of a (dog) movie star: The effect of the portrayal of dogs in movies on breed registrations in the United States

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Abstract The media is a powerful forc...

1 Attachment

PLOS ONE

The making of a (dog) movie star: The effect of the portrayal of dogs in movies on breed registrations in the United States

Samuel Weir, J. Kessler & Kessler

Department of Psychology, University of North Carolina, University of North Carolina, Chapel Hill, North Carolina, United States

10.1371/journal.pone.0261061

Abstract

The media is a powerful force that can either reflect the values of the dominant dog population. Dogs have long been in human culture and have become an integral part of our lives. The media's portrayal of dogs can influence the public's perception of dogs and their behavior. This paper examines the effect of the media's portrayal of dogs on breed registrations in the United States. We analyzed data from the American Quarter Horse Association (AQHA) and the American Paint Horse Association (APHA) from 1990 to 2018. We found that the media's portrayal of dogs has a significant effect on breed registrations. Specifically, the media's portrayal of dogs as "cute" and "friendly" has led to an increase in registrations for breeds that are perceived to be "cute" and "friendly". Conversely, the media's portrayal of dogs as "aggressive" and "dangerous" has led to a decrease in registrations for breeds that are perceived to be "aggressive" and "dangerous". These findings suggest that the media's portrayal of dogs can have a significant impact on the public's perception of dogs and their behavior. This has implications for the dog industry and for the public's understanding of dogs.

Introduction

Dogs have been used in human culture for centuries, usually to reflect human traits and emotions. (1) In recent decades, dogs have become a topic of increasing interest, and dogs have been central

